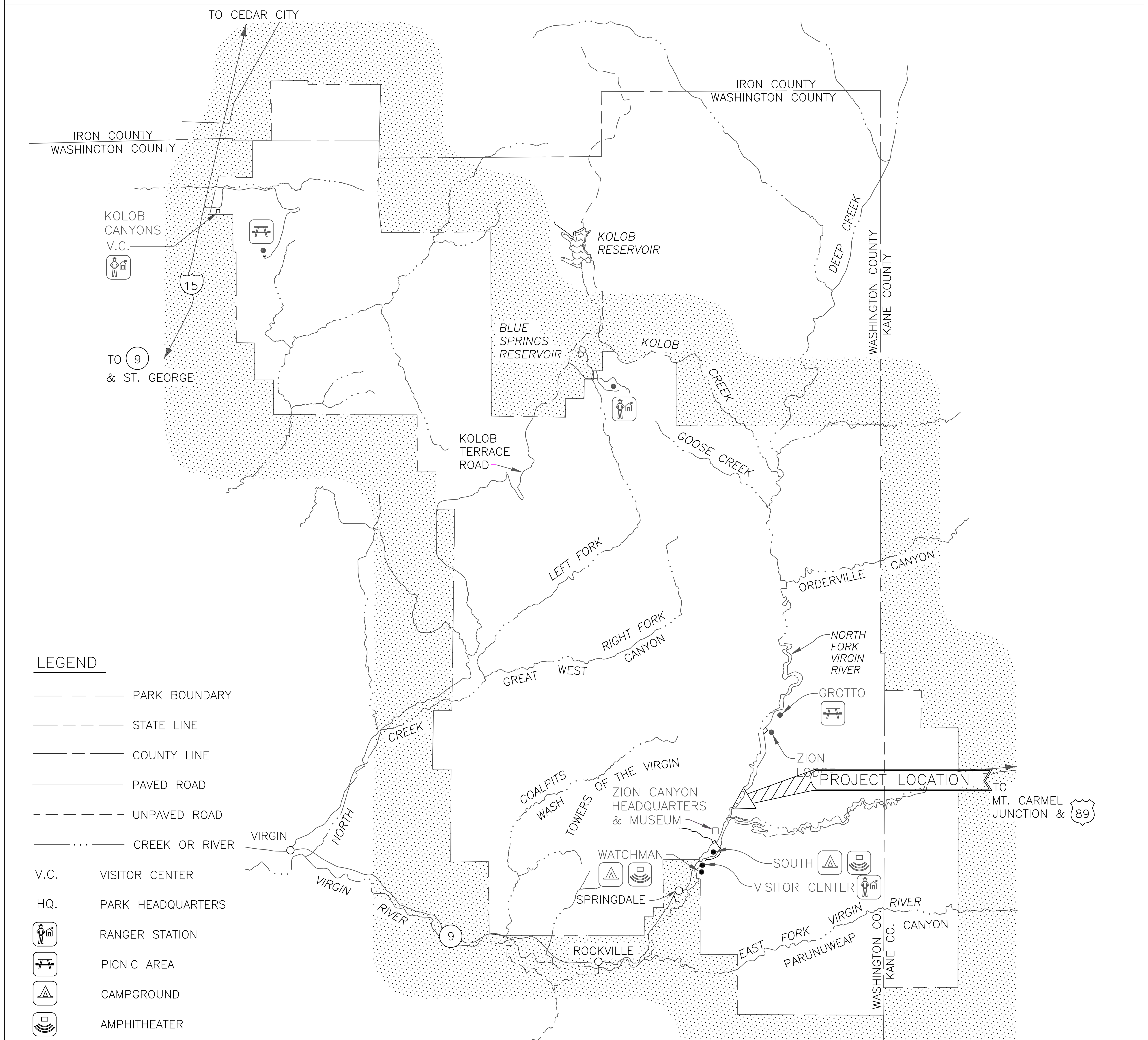

SCALE OF MILES

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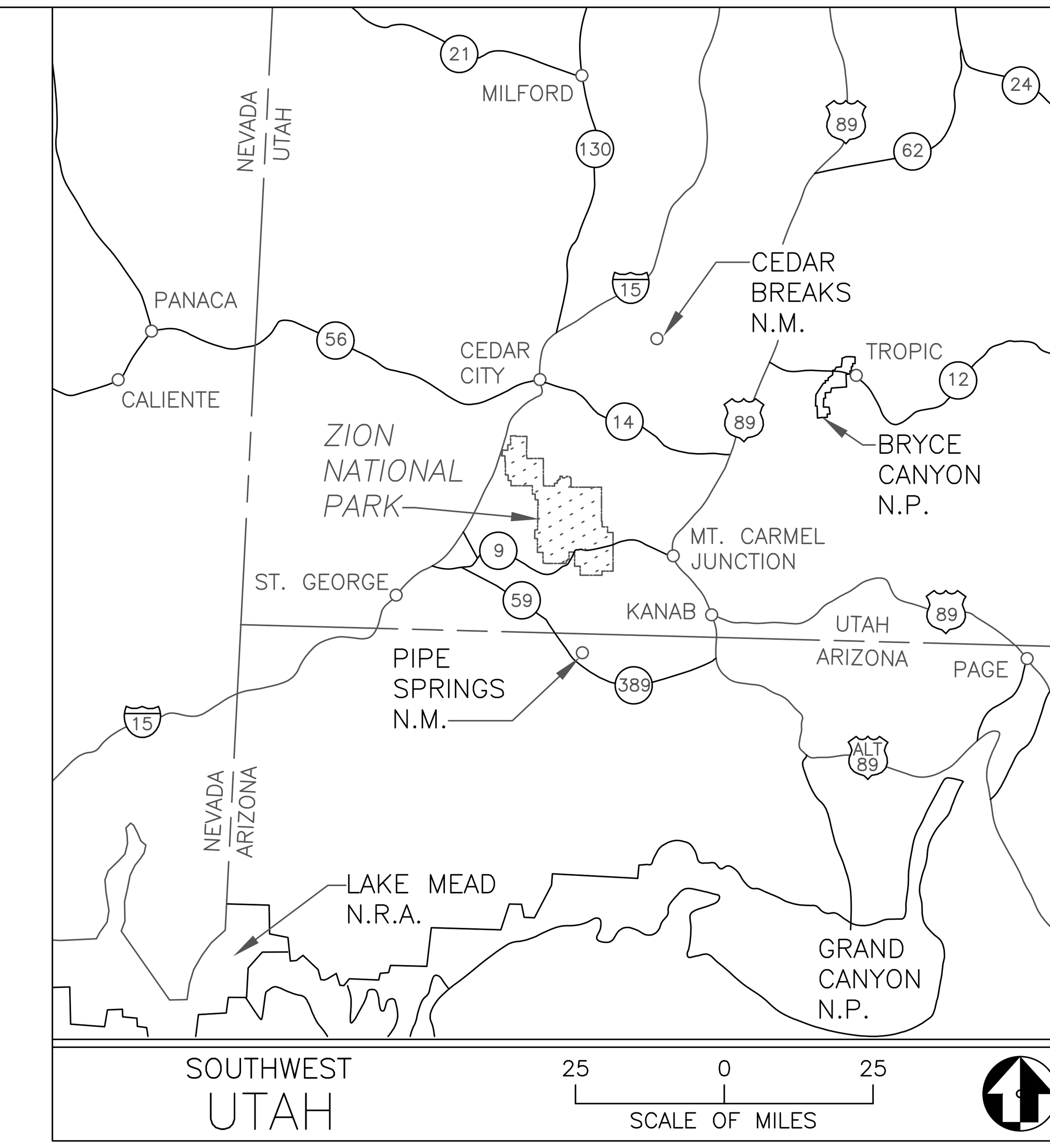
SCALE OF KILOMETERS

ZION NATIONAL PARK



NO. SHEET	SHEET TITLE
GENERAL	
CV	COVER SHEET
CIVIL	
C0	GENERAL AND CIVIL NOTES
C1	ACCESS, STAGING AND DEMOLITION
C2	EROSION CONTROL AND CONCEPTUAL TEMPORARY DIVERSION PLAN
C3	EROSION CONTROL DETAILS
C4	EROSION CONTROL NOTES
C5	OVERALL SITE PLAN
C6	INTAKE STRUCTURE AND UPPER WATERLINE PLAN
C7	INTAKE STRUCTURE
C8	FISH SCREEN DETAILS
C9	IRRIGATION CANAL AND PIPE PLAN, PROFILE AND DETAILS
C10	RIVER AND WASH PLAN
C11	RIVER AND WASH PROFILE
C12	ROCK RAMP CREST DETAILS
C13	RIVER CROSS SECTIONS (1 OF 2)
C14	RIVER CROSS SECTIONS (2 OF 2)
C15	RIVER GRADING DETAILS

ABBREVIATIONS:

ADS – ADVANCED DRAINAGE SYSTEMS
BMPS – BEST MANAGEMENT PRACTICES
CIP – CAST IN PLACE
CMP – CORRUGATED METAL PIPE
C.O.– CONTRACTING OFFICER
C.O.R – CONTRACTING OFFICER REPRESENTATIVE
EDPM – ETHYLENE, PROPYLENE, DIENE, MONOMER
EL/ELEV – ELEVATION
EW – EACH WAY
NPS – NATIONAL PARK SERVICE
NTS – NOT TO SCALE
P/N – PART NUMBER
PVC – POLYVINYL CHLORIDE
SCH – SCHEDULE
SWPPP – STORMWATER POLLUTION PREVENTION PLAN
T&B – TOP AND BOTTOM
TOW – TOP OF WALL
TYP – TYPICAL

PROFESSIONAL ENGINEER
2/1/26
11865813-2202
SUZANNE HUHTA
STATE OF UTAH

A/E FIRM – A/E CONTRACT NUMBER

PRIME:
ONEFISH ENGINEERING, LLC
PAGOSA SPRINGS, CO

Mark	Sheet	REVISION	Date	Initial
		100% SET	2/26	SBH
		90% SET (REV. 1)	1/26	SBH
		90% SET	11/25	SBH
		60% SET	7/25	SBH

QUALITY DESIGN CERTIFICATION
☐ Prepared in Accordance with Design Development (Title I) ____ Drawing No.
OR
☐ Variance from Design Development (Title I) Approved by Superintendent on ____ Date
OR
☐ Construction Drawing Not Preceded by Design Development (Title I)

Project Manager ____ Date

NATIONAL PARK SERVICE
Department of the Interior

CONSTRUCTION DRAWINGS

UNITED STATES
DEPARTMENT OF THE INTERIOR

NATIONAL PARK SERVICE
DENVER SERVICE CENTER

TITLE OF DRAWING
COVER SHEET
LOCATION WITHIN PARK
CRAWFORD DIVERSION FISH PASSAGE
NAME OF PARK
ZION NATIONAL PARK

REGION	COUNTY	STATE
INTERMOUNTAIN	WASHINGTON	UTAH

DRAWING NO. CV

PKG. NO.

SHEET 1 OF 17

GENERAL NOTES

1. THESE DRAWINGS REPRESENT THE GENERAL DESIGN INTENT TO BE IMPLEMENTED AND THE CONTRACTOR IS RESPONSIBLE FOR ALL ITEMS SHOWN ON THE PLANS. THE CONTRACTOR SHALL BE RESPONSIBLE FOR CONTACTING THE C.O. FOR ANY CLARIFICATION OR FURTHER DETAILS NECESSARY TO ACCOMMODATE ACTUAL SITE CONDITIONS. ANY DEVIATION FROM THESE PLANS WITHOUT THE C.O. APPROVAL ARE AT THE CONTRACTORS OWN RISK AND EXPENSE. THE CONTRACTOR WILL NOTIFY THE C.O. IMMEDIATELY OF ANY UNEXPECTED AND CHANGED CONDITIONS, SAFETY HAZARDS AND ENVIRONMENTAL PROBLEMS ENCOUNTERED.
2. A PRE–CONSTRUCTION MEETING WILL BE HELD AT THE JOB SITE TO BE ATTENDED BY THE NPS, ENGINEER, C.O., C.O.R., CONTRACTOR'S SUPERINTENDENT AND KEY CONTRACTOR WORK PERSONNEL. THE PURPOSE OF THE MEETING IS TO GO OVER THE WORK, PROVIDE CLARIFICATIONS, AND DISCUSS CONDITIONS OF THE PERMITS. SPECIAL ATTENTION WILL BE PAID TO FISHERIES AND ENDANGERED SPECIES ISSUES AND PROTECTION REQUIREMENTS.
3. CONTRACTOR SHALL ASSUME SOLE AND COMPLETE RESPONSIBILITY FOR SITE CONDITIONS DURING THE COURSE OF THE CONSTRUCTION OF THIS PROJECT, INCLUDING THE SAFETY OF ALL PERSONS AND PROPERTY, AND ALL ENVIRONMENTAL PROTECTION ELEMENTS. CONTRACTOR SHALL FOLLOW ALL APPLICABLE CONSTRUCTION AND SAFETY REGULATIONS. THESE REQUIREMENTS SHALL APPLY CONTINUOUSLY AND WILL NOT BE LIMITED TO NORMAL WORKING HOURS. THE CONTRACTOR SHALL DEFEND, INDEMNIFY, AND HOLD THE NATIONAL PARK SERVICE AND THE PROJECT ENGINEER HARMLESS FOR ANY AND ALL LIABILITY, REAL OR ALLEGED, IN CONNECTION WITH THE PERFORMANCE OF WORK ON THIS PROJECT, EXCEPT FROM LIABILITY ARISING FROM THE SOLE NEGLIGENCE OF THE OWNER OR ENGINEER.
4. CONTRACTOR SHALL EXERCISE CARE TO AVOID DAMAGE TO EXISTING PROPERTY, INCLUDING NATIVE TREES AND SHRUBS, PATHWAYS, FENCING AND OTHER PROPERTY IMPROVEMENTS. IF CONTRACTOR CAUSES DAMAGES TO SUCH ITEMS, CONTRACTOR SHALL BE RESPONSIBLE FOR REPAIR OR REPLACEMENT IN LIKE NUMBER, KIND, CONDITION AND SIZE AS DIRECTED BY THE OWNER.
5. THE APPROXIMATE LIMITS OF WORK ARE SHOWN ON THE DRAWINGS. EXACT LIMITS OF WORK, POINTS OF INGRESS–EGRESS, CREEK CHANNEL ACCESS, MOBILIZATION, STAGING AND WORK AREAS WILL BE FLAGGED IN THE FIELD BY THE PROJECT C.O. ALL MATERIALS, EXCESS SOIL, DEMOLITION DEBRIS AND RUBBLE, AND EQUIPMENT STORAGE MUST BE WITHIN THE STAGING AND MOBILIZATION AREA. EQUIPMENT MAINTENANCE AND REFUELING MUST OCCUR WITHIN THE STAGING AND MOBILIZATION AREAS AS DIRECTED BY APPLICABLE PERMITS. AVOID WETLAND DISTURBANCE EXCEPT WHERE DIRECT WORK IS REQUIRED.
6. CONTRACTOR IS RESPONSIBLE FOR ALL EROSION CONTROL WORK AS PART OF THE WORK. INSTALL APPROVED EROSION AND SEDIMENT CONTROL AT DOWNSTREAM END OF ALL CHANNEL GRADING AREAS PRIOR TO THE INITIATION OF CREEK GRADING WORK.
7. THE CONTRACTOR SHALL BE GIVEN COPIES OF ALL THE PERMITS, SHALL BECOME FAMILIAR WITH THE PERMIT REQUIREMENTS, AND SHALL BE RESPONSIBLE FOR ADHERENCE TO AND CONFORMANCE WITH ALL PERMIT CONDITIONS AS PART OF THE OVERALL WORK, INCLUDING POSSIBLE FINES AND MITIGATION FOR VIOLATIONS. NO ADDITIONAL CHARGES WILL BE ALLOWED FOR PERMIT COMPLIANCE WORK.
8. THE CONTRACTOR SHALL MAINTAIN AT THE SITE AT ALL TIMES ONE SIGNED COPY OF THE PROJECT DRAWINGS AND SPECIFICATIONS AND ONE COPY OF ALL REQUIRED PERMITS.
9. THE CONTRACTOR SHALL PROVIDE AND MAINTAIN ON–SITE SURVEY CONTROL.
10. CONTRACTOR IS RESPONSIBLE FOR THE LEGAL OFF–SITE DISPOSAL OF GROUT, CONCRETE, ASSOCIATED DEBRIS AND UNSUITABLE SOILS.
11. ANY NECESSARY MODIFICATIONS SHALL BE APPROVED BY THE C.O. PRIOR TO CONSTRUCTION.
12. ALL WORK SHALL BE CONTAINED ON THE SUBJECT SITE ONLY. NO STOCKPILING OR CONSTRUCTION ACTIVITY SHALL OCCUR OFF OF THE APPROVED PROJECT AREA.
13. THE CONTRACTOR SHALL PROVIDE A TEMPORARY TRASH ENCLOSURE ON SITE DURING ALL CONSTRUCTION ACTIVITIES TO CONTAIN DEBRIS AND PREVENT AIRBORNE LITTERING OFF SITE. COMPLIANCE WITH FOOD STORAGE AND WASTE DISPOSAL MUST BE ACHIEVED AT ALL TIMES. THE PROJECT SITE WILL BE CLEANED UP AT THE END OF EACH WORK PERIOD DAILY TO REDUCE ATTRACTION OF WILDLIFE.
14. A WATER TRUCK SHALL BE USED DAILY ON SITE TO WET DOWN ALL EARTHWORK AND TO CONTROL AIRBORNE PARTICLES.
15. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE LOCATION OF AND PROTECTION OF ALL EXISTING UNDERGROUND UTILITIES DURING CONSTRUCTION, OTHER THAN THOSE SPECIFIED IN THE DEMO PLAN.
16. PROJECTS SHALL SUBMIT A DUST CONTROL PLAN WITH DETAILS ON EQUIPMENT, SCHEDULING AND REPORTING OF DUST CONTROL ACTIVITIES.
17. NO CHEMICAL STORAGE, WASHING OF PAINT OR OTHER CHEMICALS IS ALLOWED WITHIN THE CRITICAL ROOT ZONE OR IN THE RIVER. NO CLEANING, SPILLING, OR SPRAYING OF PAINT OR CHEMICALS ANYWHERE ON THE EARTH SURFACE OF THE PROJECT SITE, OR IN THE RIVER, IS ALLOWED. USE OF PAINT TO MARK THE SITE FOR GROUND WORK IS THE ONLY EXCEPTION TO THIS REQUIREMENT.
18. RE–FUELING SHALL BE PERFORMED OUTSIDE OF THE 100–YEAR FLOODPLAIN; SPILL CONTAINMENT KITS SHALL BE AVAILABLE ONSITE.
19. CONTRACTOR SHALL PROVIDE A TRAFFIC CONTROL PLAN AND CONSTRUCTION SCHEDULE TO ADDRESS ANY CONSTRUCTION IMPACTS TO EXISTING ROADWAYS.
20. CONTRACTOR SHALL SUBMIT AN ACCIDENT PREVENTION PLAN.
21. WORK TO BE PERFORMED DURING DAYTIME/DAYLIGHT HOURS.

MOBILIZATION AND STAGING

1. CONTRACTOR SHALL SUBMIT AN ACCESS, STAGING AND STOCKPILE PLAN FOR APPROVAL NO LESS THAN 10 WORKING DAYS PRIOR TO PRE–CONSTRUCTION MEETING.
2. CONTRACTORS SHALL SUBMIT SAFETY PLAN FOR APPROVAL NO LESS THAN 10 WORKING DAYS PRIOR TO PRE–CONSTRUCTION MEETING.
3. THE CONTRACTOR SHALL KEEP THE WORK AREAS IN NEAT CONDITION FREE OF DEBRIS AND LITTER FOR THE DURATION OF THE PROJECT.
4. ALL EQUIPMENT SHALL BE WASHED PRIOR TO MOBILIZATION TO THE SITE TO MINIMIZE THE INTRODUCTION OF FOREIGN MATERIALS TO THE PROJECT SITE. ALL EQUIPMENT SHALL BE FREE OF OIL, HYDRAULIC FLUID, AND DIESEL FUEL LEAKS. TO PREVENT THE INTRODUCTION OF NOXIOUS WEEDS ALL EQUIPMENT SHALL BE POWER WASHED OR CLEANED TO REMOVE MUD AND SOIL PRIOR TO MOBILIZATION INTO THE PARK.
5. CONTRACTOR SHALL COORDINATE DATE AND TIME OF NPS COR INSPECTION OF ALL CONSTRUCTION EQUIPMENT IN THE SOUTH ENTRANCE MONUMENT PARKING AREA NO LESS THAN TWO WORKING DAYS PRIOR TO THE ARRIVAL OF EQUIPMENT.
6. THE USE OF CANOLA OIL OR OTHER BIODEGRADABLE FUELS AND FLUIDS IS RECOMMENDED WHEN WORKING IN SENSITIVE RIVERINE ENVIRONMENTS.

DEWATERING

1. THE CONTRACTOR SHALL BE RESPONSIBLE FOR DEVELOPING AND COMPLYING WITH A DEWATERING PLAN NO LESS THAN 10 WORKING DAYS PRIOR TO THE PRE–CONSTRUCTION MEETING. CONTRACTOR IS RESPONSIBLE FOR REMOVAL AND DISPOSITION OF ALL WATER CONTROL STRUCTURES.
2. THE CONTRACTOR SHALL FURNISH, INSTALL, AND OPERATE ALL NECESSARY MACHINERY, APPLIANCES, AND EQUIPMENT TO DIVERT FLOWING WATER AROUND WORK AREAS, AND TO KEEP EXCAVATIONS AND TRENCHES FREE FROM WATER DURING CONSTRUCTION AT NO ADDITIONAL EXPENSE TO THE GOVERNMENT. CONTRACTOR SHALL DISPOSE OF THE WATER SO AS NOT TO CAUSE INJURY TO PROPERTY, OR TO CAUSE A NUISANCE OR A MENACE TO THE PUBLIC, OR TO DEGRADE WATER QUALITY.
3. TURBID DEWATERING FLOWS SHALL BE PUMPED INTO A HOLDING FACILITY. AT NO TIME SHALL TURBID WATER BE ALLOWED BACK INTO THE STREAM CHANNEL UNTIL WATER IS CLEAR OF SILT OR FUEL THAT DRIPS OR SPILLS INTO THE CHANNEL.

DEMOLITION NOTES

1. ALL EXISTING FEATURES WITH IN THE CONSTRUCTION AREA TO REMAIN ARE TO BE PROTECTED IN PLACE DURING DEMOLITION AND CONSTRUCTION OPERATIONS.
2. CONTRACTOR SHALL GRIND, REMOVE, AND DISPOSE OF STUMPS WITHIN ACCESS ROAD ALIGNMENT AT NO ADDITIONAL EXPENSE TO THE GOV'T.

GRADING NOTES

1. PROPOSED GRADES MAY REQUIRE ADJUSTMENT BASED ON SITE CONDITIONS DURING CONSTRUCTION. ALL FIELD CHANGES ARE TO BE APPROVED BY THE C.O.
2. SOIL ROCK OR OTHER MATERIALS TO BE USED FOR BACKFILL, CHANNEL ARMORING, AND ROCK VANES SHALL BE APPROVED BY THE C.O.
3. ALL FILL MATERIAL SHALL BE COMPACTED.
4. CONTRACTOR SHALL MINIMIZE DISTURBANCE TO EXISTING GROUND OUTSIDE OF GRADING LIMITS. SLOPES SHALL GENERALLY MATCH EXISTING TERRAIN WHERE POSSIBLE.
5. ALL SHOULDERING/DAYLIGHT GRADING SHALL HAVE A MAXIMUM SLOPE OF 3 HORIZONTAL : 1 VERTICAL.
6. ALL EXCAVATIONS AND GRADING SHALL BE IN ACCORDANCE WITH APPENDIX 33 OF THE "UNIFORM BUILDING CODE," 1994 EDITION FOR FEE SCHEDULES, AND APPENDIX K OF THE "UTAH UNIFORM BUILDING STANDARD ACT RULES" FOR GRADING, EXCAVATION AND EARTHWORK CONSTRUCTION.
7. THE CONTRACTOR SHALL PROVIDE SUITABLE EQUIPMENT TO CONTROL DUST AND AIR POLLUTION CAUSED BY CONSTRUCTION OPERATIONS. THE CONTRACTOR SHALL ALSO PROVIDE SUITABLE MUD AND DIRT CONTAINMENT TO MAINTAIN THE WORK SITE, ACCESS ROADWAYS AND ADJACENT PROPERTIES IN A CLEAN CONDITION.
8. ALL IMPORTED STRUCTURAL FILL SHALL BE APPROVED BY THE C.O. PRIOR TO DELIVERY TO THE SITE. ALL STRUCTURAL FILL SHALL BE PLACED IN 8 INCH LOOSE HORIZONTAL LIFTS AND COMPACTED TO A MINIMUM OF 95 PERCENT OF MAXIMUM DRY DENSITY (ASTM D–1557).
9. NO WORK, EQUIPMENT, OR MATERIALS SHALL BE PERMITTED OUTSIDE THE DESIGNATED CONSTRUCTION LIMITS AS SHOWN ON THESE PLANS.

PROFESSIONAL ENGINEER
2/1/26
\# 11865813-2202
SUZANNE
HUHTA
STATE OF UTAH

DESIGNED: SBH CADD SBH TECH. REVIEW: CAH DATE: 02/01/26	SUB SHEET NO. C0	TITLE OF SHEET GENERAL AND CIVIL NOTES CRAWFORD DIVERSION FISH PASSAGE ZION NATIONAL PARK	DRAWING NO. ——— PMIS/PKG NO. SHEET 2 OF 17

ACCESS ROUTE, STAGING AREA, AND DEMOLITION

N

0 40 80 160

SCALE 1" = 80'
(IF PRINTED 34"X22")

TRAFFIC NOTES:
1. CONTRACTOR SHALL SUBMIT AN ACCIDENT PREVENTION PLAN FOR PARK REVIEW AND APPROVAL. CONTRACTOR SHALL COMPLY WITH TRAFFIC CONTROL IN ACCIDENT PREVENTION PLAN.
2. CONTRACTOR TO PROVIDE TRAFFIC GUARD WHEN EQUIPMENT ENTERS/EXITS ZION–MT CARMEL HIGHWAY.
3. EQUIPMENT TO FOLLOW THE EXISTING TRAFFIC DIRECTION.
4. FENCE CLOSURES MUST BE SECURED AT ALL TIMES.
5. CONTRACTOR PARKING ALONG ACCESS ROAD AND IN STAGING AREAS.

DEMOLITION NOTES:
1. DEMOLISH, HAUL AND DISPOSE OF MATERIAL OFF SITE.

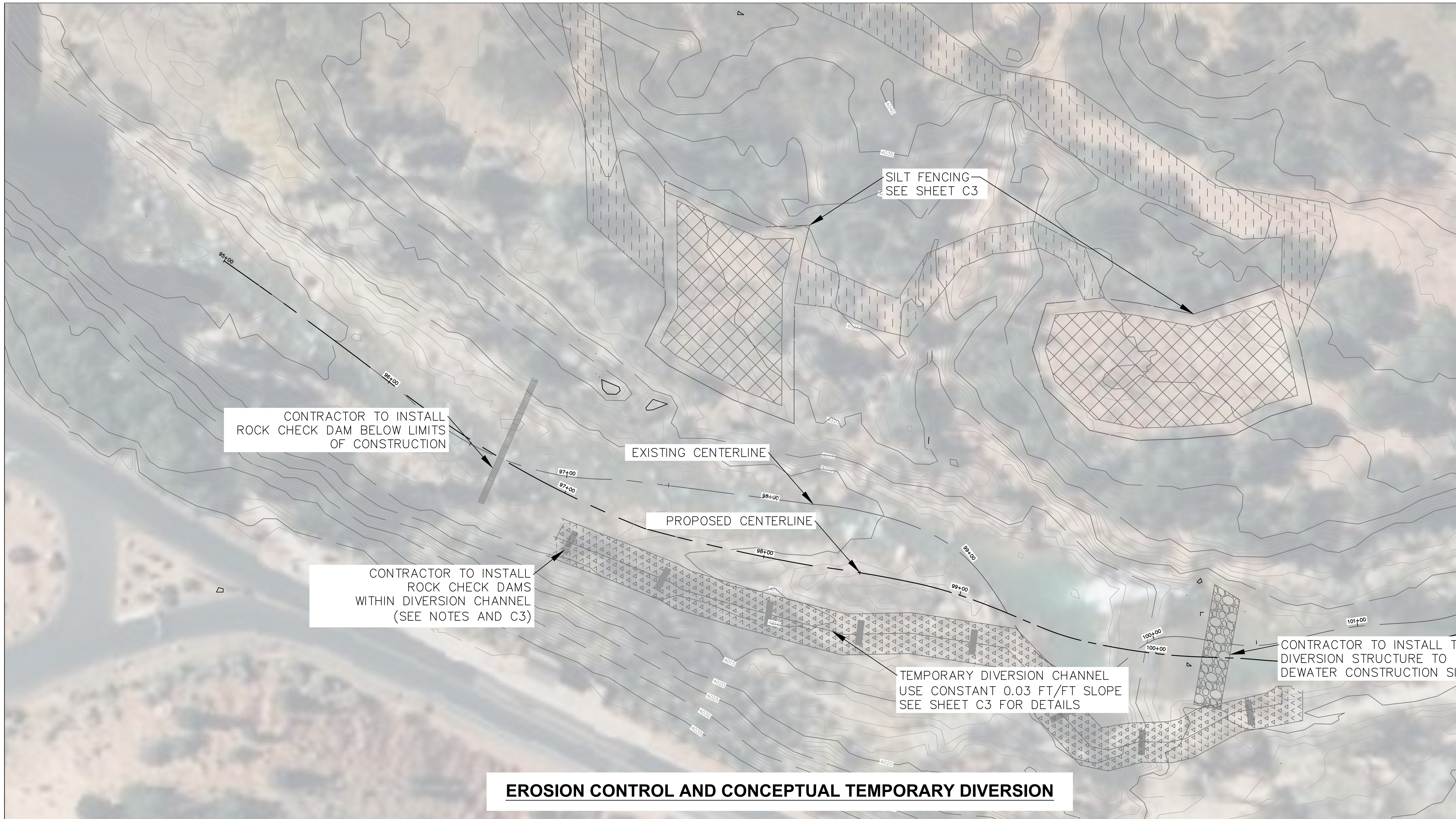
GENERAL NOTES:
1. AT CONCLUSION OF WORK, ALL STAGING AREAS SHALL BE RETURNED TO ORIGINAL ELEVATIONS AND SCARIFIED FOR SEEDING. SEEDING WILL BE DONE BY NPS.
2. ALTERNATIVE ACCESS ROAD (LOWER ACCESS ROAD) SHALL BE GRADED TO PRE–PROJECT ELEVATIONS.

EXAMPLE OF EXISTING GROUTED RIPRAP LOCATED ADJACENT TO THE DAM

PROFESSIONAL ENGINEER
2/1/26
11865813-2202
SUZANNE HUHTA
STATE OF UTAH

DESIGNED: SBH	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
CADD SBH	C1	ACCESS, STAGING AND DEMOLITION	PMIS/PKG NO.
TECH. REVIEW: CAH		CRAWFORD DIVERSION FISH PASSAGE ZION NATIONAL PARK	SHEET 3 OF 17
DATE: 02/01/26			

FILE NAME: 2026-02 CRAWFORD ACCESS C1.DWG
PLOT TIME/DATE: 1:29 PM, 2/1/2026
PLOTTED BY: Suzanne Huhta
PLOT STYLE: Acad.ctb



EROSION CONTROL AND CONCEPTUAL TEMPORARY DIVERSION

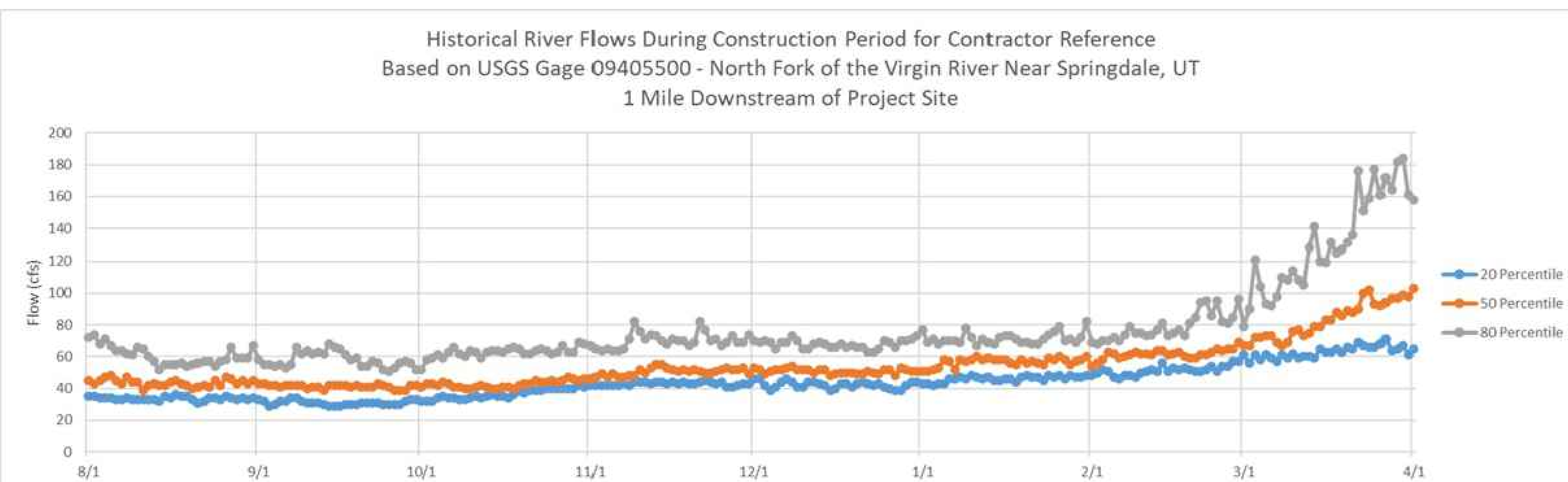
SCALE 1" = 30'
(IF PRINTED 34"X22")

CARE OF WATER NOTES:

1. THE CONTRACTOR SHALL BE RESPONSIBLE FOR COMPLIANCE WITH ALL LOCAL, STATE, AND FEDERAL REGULATIONS AND BEST MANAGEMENT PRACTICES (BMPS).
2. CONTRACTOR SHALL DEVELOP AND SUBMIT A CARE OF WATER PLAN TO C.O. FOR APPROVAL.
3. PLAN PRESENTED HERE FOR ILLUSTRATIVE PURPOSES. CONTRACTOR SHALL BE RESPONSIBLE FOR FINAL CARE OF WATER PLAN.
4. SPLIT CHANNEL DEWATERING/CARE OF WATER IS CONSIDERED A VIABLE OPTION, COMBINATION OF CONCEPTUAL PLAN AND SPLIT CHANNEL IS ALSO A VIABLE OPTION.
5. DIVERSION CHANNEL CARE OF WATER OPTIONS SHALL USE ROCK CHECK STRUCTURES EXTENDING FULLY ACROSS DIVERSION CHANNEL BOTTOM.
   - 5.1. ROCK CHECK DAM SPACING SHALL BE 50' MAXIMUM.
   - 5.2. DIVERSION CHANNEL SHALL BE INSPECTED DAILY TO ASSURE PROPER FUNCTION.
   - 5.3. SEDIMENT BEHIND ROCK CHECK DAMS TO BE CLEARED WHEN BURIED 1/3 OF DEPTH.
6. AREAS DISTURBED BY CARE OF WATER WILL BE RETURNED TO ORIGINAL CONDITION OR BETTER.
7. ALL TEMPORARY DIVERSION MATERIALS MUST BE NON-ERODIBLE.
8. CONTRACTOR SHALL BE RESPONSIBLE FOR DEVELOPMENT OF STAGING AREAS, ACCESS AND CARE OF WATER DURING CONSTRUCTION.

EROSION CONTROL NOTES:

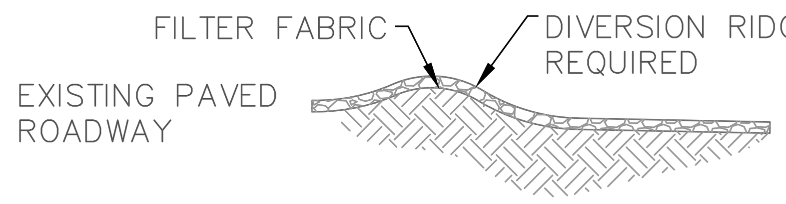
1. ~280' (LOWER STAGING AREA) AND ~400' (UPPER STAGING AREA) SILT FENCING PERIMETER TO BE INSTALLED A MINIMUM OF 6' OFFSET FROM TOE OF STOCKPILED MATERIAL AND WHERE DIRECTED BY C.O. (SEE DETAIL B, SHEET C3)



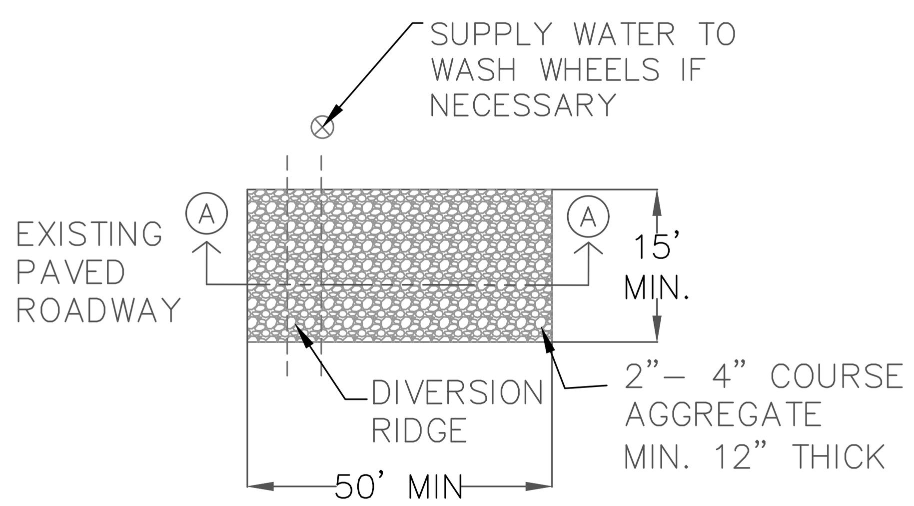
HISTORICAL FLOW PROVIDED FOR CONTRACTOR USE IN SIZING TEMPORARY DIVERSION

PROFESSIONAL ENGINEER
11865813-2202
SUZANNE HUHTA
STATE OF UTAH

DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
SBH	C2	EROSION CONTROL AND CONCEPTUAL TEMPORARY DIVERSION PLAN	———
CADD: SBH		CRAWFORD DIVERSION FISH PASSAGE	PMIS/PKG NO.
TECH. REVIEW: CAH		ZION NATIONAL PARK	SHEET
DATE: 02/01/26			4 OF 16



SECTION A–A



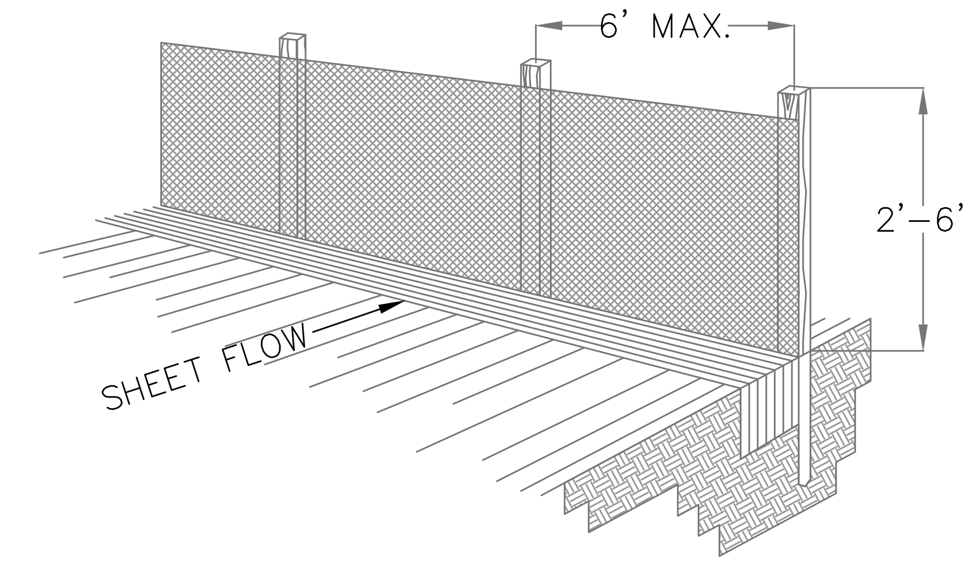
PLAN VIEW

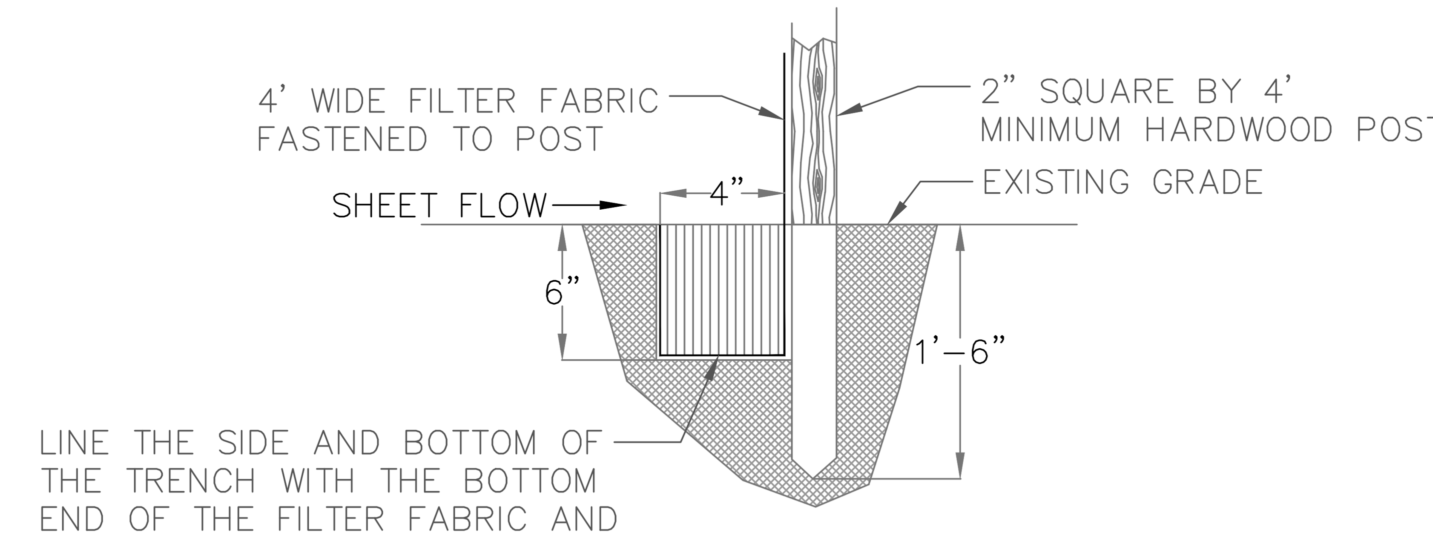
1. MAINTAIN THE GRAVEL PAD IN A CONDITION TO PREVENT MUD OR SEDIMENT FROM LEAVING THE CONSTRUCTION SITE. REPLACE GRAVEL MATERIAL WHEN SURFACE VOIDS ARE VISIBLE. AFTER EACH RAINFALL, INSPECT ANY STRUCTURE USED TO TRAP SEDIMENT AND CLEAN IT OUT AS NECESSARY. IMMEDIATELY REMOVE ALL OBJECTIONABLE MATERIALS SPILLED, WASHED, OR TRACKED ONTO PUBLIC ROADWAYS. REMOVE ALL SEDIMENT DEPOSITED ON PAVED ROADWAYS WITHIN 24 HOURS.
2. ROUTINE WASHING OF THE TRACK PAD IS REQUIRED.

CONSTRUCTION SPECIFICATIONS:

1. THE AGGREGATE SIZE FOR CONSTRUCTION OF THE PAD SHALL BE 2 TO 4 INCH STONE. PLACE GRAVEL TO THE SPECIFIC GRADE AND DIMENSIONS SHOWN ON THE PLANS, AND SMOOTH IT. USE GEOTEXTILE FABRICS, IF NECESSARY, TO IMPROVE STABILITY OF THE FOUNDATION IN LOCATIONS SUBJECT TO SEEPAGE OR HIGH WATER TABLE. THE WIDTH OF THE PAD SHALL NOT BE LESS THAN THE FULL WIDTH OF ALL POINTS OF INGRESS OR EGRESS AND IN ANY CASE SHALL NOT BE LESS THAN 15 FEET WIDE. THE LENGTH OF THE PAD SHALL BE AS REQUIRED, BUT NOT LESS THAN 50 FEET.
2. LOCATE CONSTRUCTION ENTRANCES AND EXITS TO LIMIT SEDIMENT LEAVING THE SITE AND TO PROVIDE FOR MAXIMUM UTILITY BY ALL CONSTRUCTION VEHICLES. AVOID ENTRANCES WHICH HAVE STEEP GRADES. THE ENTRANCE SHALL BE MAINTAINED IN A CONDITION THAT WILL PREVENT TRACKING OR FLOWING OF SEDIMENT ONTO PUBLIC RIGHT–OF–WAYS. THIS MAY REQUIRE PERIODIC TOP DRESSING WITH ADDITIONAL STONE AS CONDITIONS DEMAND, AND REPAIR AND/OR CLEANOUT OF ANY MEASURES USED TO TRAP SEDIMENT. ALL SEDIMENT SPILLED, DROPPED, WASHED, OR TRACKED ONTO PUBLIC RIGHT–OF–WAY SHALL BE REMOVED WITHIN 24 HOURS. PROVIDE DRAINAGE TO CARRY WATER TO A SEDIMENT TRAP OR OTHER SUITABLE OUTLET. WHEN NECESSARY, WHEELS SHALL BE CLEANED TO REMOVE SEDIMENT PRIOR TO ENTRANCE ONTO PUBLIC RIGHT–OF–WAY. WHEN WASHING IS REQUIRED, IT SHALL BE DONE ON AN AREA STABILIZED WITH CRUSHED STONE THAT DRAINS INTO AN APPROVED SEDIMENT TRAP OR SEDIMENT BASIN. ALL SEDIMENT SHALL BE PREVENTED FROM ENTERING ANY STORM DRAIN, DITCH OR WATERCOURSE THROUGH USE OF SAND BAGS, GRAVEL, OR OTHER APPROVED METHODS.
3. PREFABRICATED TRACK CONTROL MAT MAY BE USED IN PLACE OF STONE PAD IF APPROVED BY THE C.O.

A/— STABILIZED CONSTRUCTION EXIT
SCALE: N.T.S.



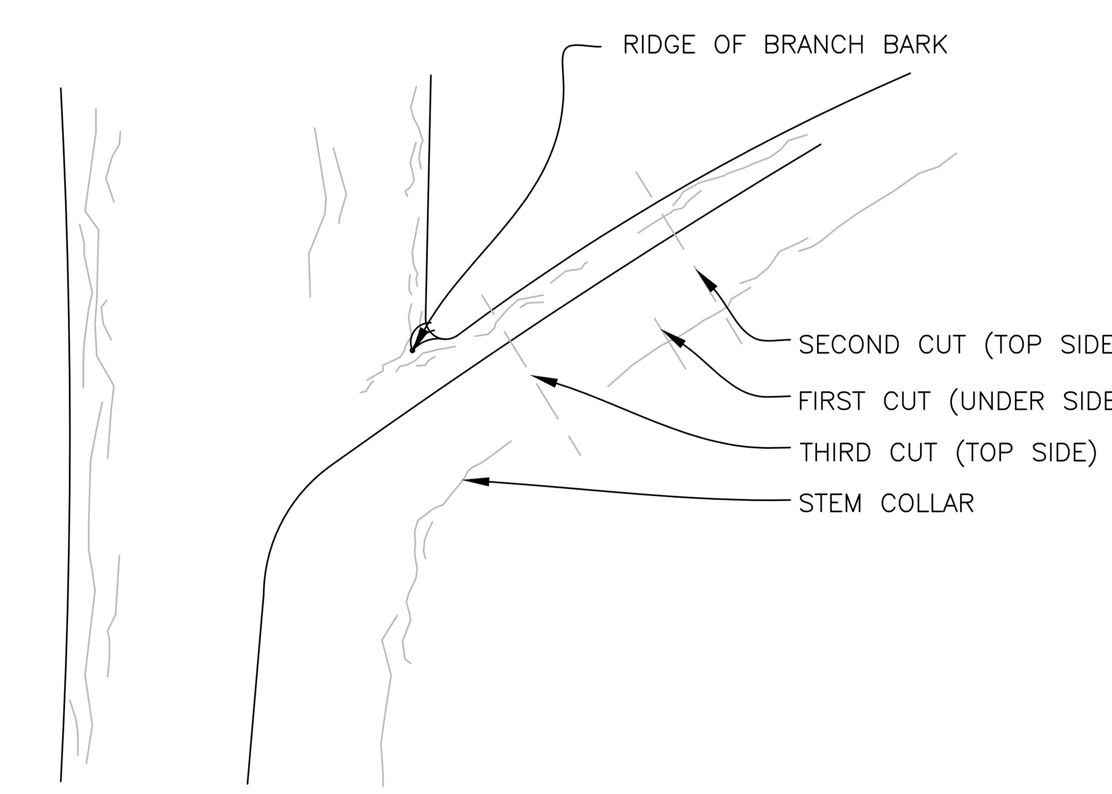


SECTION

B/— TEMPORARY SILT FENCE
SCALE: N.T.S.

NOTES:

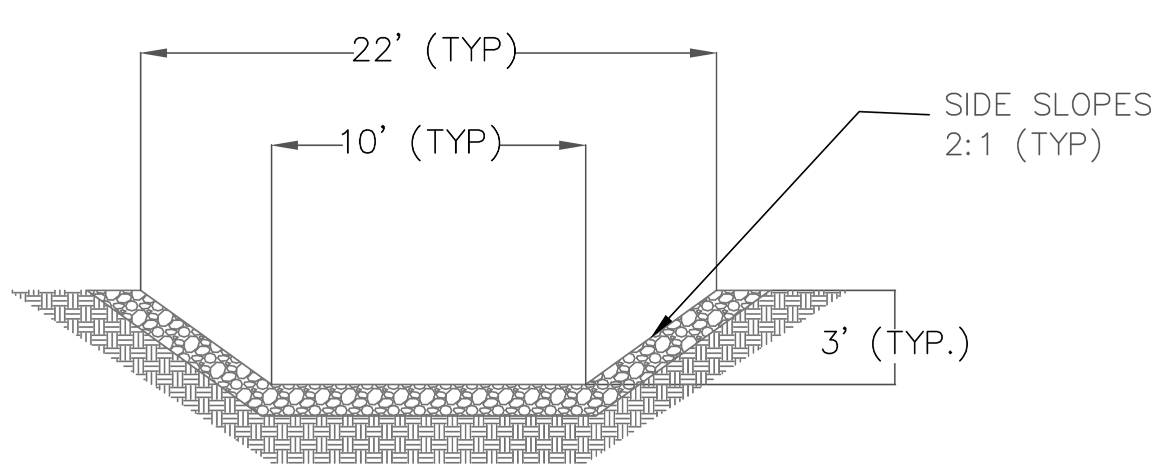
1. WHERE POSSIBLE, LAYOUT THE SILT FENCE 5' TO 10' BEYOND THE TOE OF SLOPE.
2. ALIGN THE FENCE ALONG THE CONTOUR AS CLOSE AS POSSIBLE.
3. WHEN EXCAVATING THE TRENCH, USE MACHINERY THAT WILL PRODUCE NO MORE THAN THE DESIRED DIMENSIONS.
4. EXTEND THE BOTTOM 1'–4" OF FILTER FABRIC TO LINE ALL THREE SIDES OF THE TRENCH.
5. TO AVOID EXCESSIVE PONDING OF WATER AT LOW POINTS ALONG THE FENCE, PROVIDE AN OPENING IN THE SILT FENCE AND INSTALL A CHECK DAM.
6. AVOID USING JOINTS ALONG THE FENCE AS MUCH AS POSSIBLE. IF A JOINT IS NECESSARY, SPLICE THE FILTER FABRIC AT A POST WITH 6" OVERLAPS AND SECURELY FASTEN BOTH ENDS TO THE POST.
7. MAINTAIN A PROPERLY FUNCTIONING SILT FENCE THROUGHOUT THE DURATION OF THE PROJECT OR UNTIL DISTURBED AREAS HAVE BEEN VEGETATED.
8. REMOVE SEDIMENT AS IT ACCUMULATES AND PLACE IT IN A STABLE AREA APPROVED BY THE C.O.
9. STRAW WATTLES MAY BE USED IN LIEU OF SILT FENCE AT THE DISCRETION OF THE C.O.



NOTES:

1. IDENTIFY BRANCHES THAT NEED TO BE REMOVED AND OBTAIN APPROVAL FROM C.O.
2. FOR EACH BRANCH, LOCATE THE STEM COLLAR AND THE RIDGE OF BARK FORMED ABOVE THE BRANCH UNION. THE STEM COLLAR IS USUALLY EVIDENCED BY A SLIGHT SWELLING AT THE UNION.
3. NEVER CUT THE BRANCH FLUSH WITH THE TRUNK; ALWAYS CUT OUTSIDE OF THE COLLAR AND BARK RIDGE.
4. FOR ALL BRANCHES 1" AND GREATER USE THE THREE CUT METHOD TO AVOID TEARING THE BARK IN THE COLLAR AREA.
5. THIS DETAIL IS SHOWN FOR INFORMATION ONLY IF TREE PRUNING IS REQUIRED. CONTRACTOR SHALL COORDINATE WITH THE C.O. PRIOR TO TREE PRUNING.

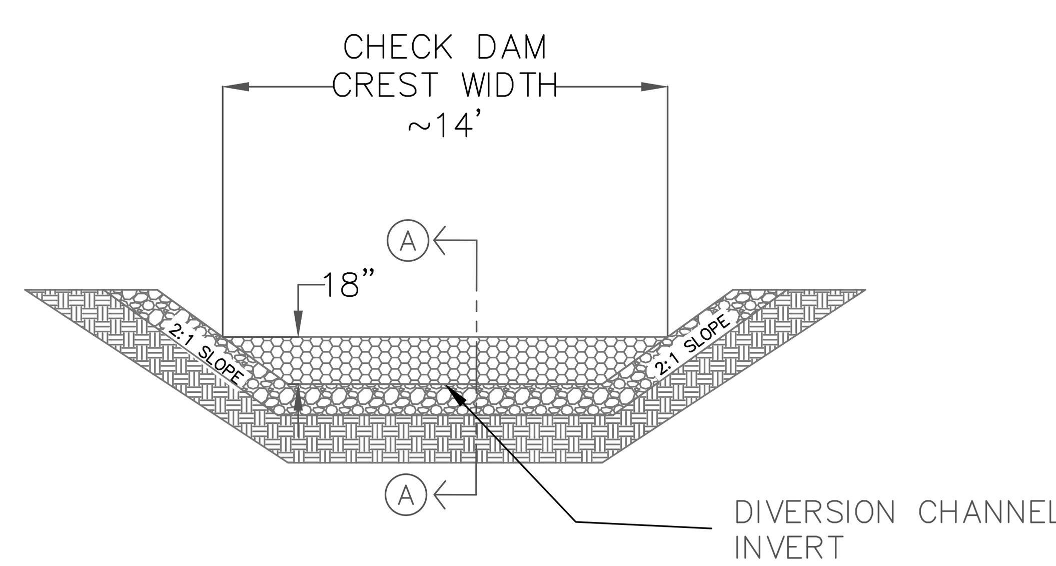
C/— TREE CUT PRUNING METHOD
SCALE: N.T.S.



NOTES:

1. CHANNEL SLOPE TO MATCH THE STARTING AND ENDING ELEVATIONS.
2. BACKFILL CHANNEL AND RESTORE SURFACE TO PREVIOUS ELEVATIONS AFTER PROJECT COMPLETION.
3. ROCK LINING 1' THICK.
4. D_{50} = 6"ø
5. ALL MATERIALS TO BE NON–ERODIBLE.

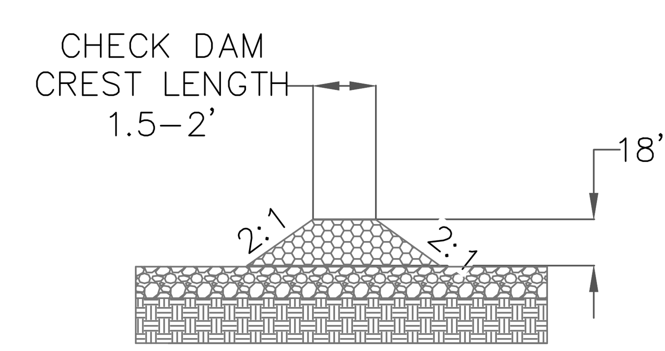
D/— TEMPORARY DIVERSION CHANNEL (TYP.)
SCALE: N.T.S.



NOTES:

1. CHECK DAMS EVERY 50 FT.
2. D_{50} = 12"ø

E/— ROCK CHECK DAM PLAN VIEW (TYP.)
SCALE: N.T.S.



SECTION A–A
SCALE: N.T.S.

DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
SBH	C3	EROSION CONTROL DETAILS	——
CADD: SBH			PMIS/PKG NO.
TECH. REVIEW: CAH		CRAWFORD DIVERSION FISH PASSAGE	SHEET
DATE: 02/01/26		ZION NATIONAL PARK	5 OF 17

PROFESSIONAL ENGINEER 2/1/26 # 11865813-2202 SUZANNE HUHTA STATE OF UTAH

FILE NAME: 2026–02 CRAWFORD EROSION CONTROL DETAILS C3 C4.DWG
PLOT TIME/DATE: 1:29 PM, 2/1/2026
PLOTTED BY: Suzanne Huhta
PLOT STYLE: Acad.ctb

EROSION & SEDIMENTATION CONTROL NOTES:

1. AT ALL TIMES DURING CONSTRUCTION, THE CONTRACTOR SHALL BE RESPONSIBLE FOR PREVENTING AND CONTROLLING EROSION DUE TO WIND AND RUNOFF. THE CONTRACTOR SHALL ALSO BE RESPONSIBLE FOR MAINTAINING EROSION CONTROL FACILITIES SHOWN.

2. THE CONTRACTOR SHALL REMOVE LITTER, CONSTRUCTION DEBRIS, AND CONSTRUCTION CHEMICALS ON A DAILY BASIS, AND, PRIOR TO ANY ANTICIPATED STORM EVENT; OR OTHERWISE PREVENT SUCH MATERIAL FROM BECOMING A POLLUTANT SOURCE.

3. CONTRACTOR SHALL USE VEHICLE TRACKING CONTROL AT ALL LOCATIONS WHERE VEHICLE WILL ENTER OR EXIT THE SITE. VEHICLE TRACKING CONTROL FACILITIES, SILT FENCE, IF APPLICABLE, AND INLET PROTECTION WILL BE MAINTAINED WHILE CONSTRUCTION IS IN PROGRESS, MOVED WHEN NECESSARY FOR PHASING OF THE WORK , UNTIL ALL WORK IS COMPLETE.

4. THE CONTRACTOR SHALL BE RESPONSIBLE FOR KEEPING ROADS CLEAN OF DEBRIS FROM TRAFFIC FROM THE SITE. IF SEDIMENT ESCAPES THE CONSTRUCTION SITE, THE CONTRACTOR SHALL REMOVE OFF–SITE ACCUMULATIONS OF SEDIMENT IMMEDIATELY TO MINIMIZE OFFSITE IMPACTS. AT NO TIME SHALL SEDIMENT BE WASHED DOWN INTO THE RIVER DRAINAGES.

5. CONTRACTOR SHALL BE RESPONSIBLE FOR CLEANING DRAINAGE AND EROSION CONTROL FACILITIES AS REQUIRED, AND SHALL REMOVE SEDIMENT FROM SEDIMENT TRAPS OR PONDS WHEN THE DESIGN CAPACITY HAS BEEN REDUCED BY 50%. ALL EROSION CONTROL DEVICES SHALL BE INSPECTED DAILY AND PROPERLY MAINTAINED UNTIL FINAL SITE STABILIZATION IS ACHIEVED.

6. THE CONTRACTOR SHALL INITIATE STABILIZATION MEASURES AS SOON AS PRACTICABLE IN PORTIONS OF THE SITE WHERE CONSTRUCTION ACTIVITIES HAVE TEMPORARILY OR PERMANENTLY CEASED, BUT IN NO CASE MORE THAN 14 DAYS AFTER THE CONSTRUCTION ACTIVITY IN THAT PORTION OF THE SITE HAS TEMPORARILY OR PERMANENTLY CEASED.

7. ADDITIONAL EROSION CONTROL MEASURES MAY BE REQUIRED DUE TO UNFORESEEN PROBLEMS OR IF THE PLAN DOES NOT FUNCTION AS INTENDED.

8. THE CONTRACTOR SHALL KEEP A RECORD OF THE DATES WHEN MAJOR GRADING ACTIVITIES OCCUR, WHEN CONSTRUCTION ACTIVITIES TEMPORARILY OR PERMANENTLY CEASE ON A PORTION OF THE SITE, AND WHEN STABILIZATION MEASURES ARE INITIATED.

9. DUST CONTROL–CONSTRUCTION TRAFFIC MUST ENTER AND EXIT THE SITE AT THE STABILIZED CONSTRUCTION EXIT. THE PURPOSE IS TO TRAP DUST AND MUD THAT WOULD OTHERWISE BE CARRIED OFF–SITE BY CONSTRUCTION TRAFFIC.

10. WATER TRUCKS WILL BE USED AS NEEDED DURING CONSTRUCTION TO REDUCE DUST GENERATED ON THE SITE. DUST CONTROL MUST BE PROVIDED BY THE CONTRACTOR TO A DEGREE THAT IS ACCEPTABLE TO THE C.O., AND IN COMPLIANCE WITH APPLICABLE LOCAL AND STATE DUST CONTROL REGULATIONS.

11. LOCATIONS WHERE VEHICLES ENTER AND EXIT THE SITE MUST BE INSPECTED FOR EVIDENCE OF OFF–SITE SEDIMENT TRACKING. A STABILIZED CONSTRUCTION EXIT WILL BE CONSTRUCTED WHERE VEHICLES EXIT AND ENTER THE PROJECT. THIS EXIT WILL BE MAINTAINED OR SUPPLEMENTED AS NECESSARY TO PREVENT SEDIMENT FORM LEAVING THE SITE ON VEHICLES.

12. ALL EROSION CONTROL SHALL BE 100% BIODEGRADABLE AND WILDLIFE FRIENDLY AND CONTAIN NO SYNTHETIC STRINGS/CORDS/NETTING THAT CAN CAUSE ENTANGLEMENT OR BE HARMFUL TO DIGESTION BY WILDLIFE.

13. ALL EROSION CONTROL SHALL BE CERTIFIED WEED–FREE – SEE SPECIFICATIONS.

14. DISTURBANCES OF THE RIPARIAN ZONE MUST BE LIMITED TO INDICATED ACCESS POINTS PRIOR TO THE OPERATION OF HEAVY EQUIPMENT (DOZERS, CRANES, TRUCKS).

15. THE FASTENING OF ROPES, CABLES, OR FENCING TO TREES IS PROHIBITED.

16. TO ENSURE BANK STABILITY, TREES REMOVED WITHIN 15 FEET OF THE TOP OF THE RIVER BANK SHALL BE CUT FLUSH TO THE GROUND; STUMPS AND ROOTS SHALL BE LEFT IN PLACE; INDISCRIMINATE BULLDOZING OF RIPARIAN TREES IS PROHIBITED. C.O. APPROVAL IS REQUIRED FOR ANY AND ALL TREE REMOVAL.

17. LITTER AND CONSTRUCTION DEBRIS SHALL BE CONTAINED DAILY. CONSTRUCTION AND/OR WORKER GENERATED DEBRIS/GARBAGE MUST BE CONTAINED ON SITE IN PROPERLY COVERED CONTAINERS; DAILY CLEAN UP MUST BE PERFORMED TO PREVENT MATERIALS FROM ENTERING THE RIVER.

18. ALL CONSTRUCTION EQUIPMENT MUST BE INSPECTED DAILY FOR HYDRAULIC AND FUEL LEAKS, AND REPAIRED AS NECESSARY. WHEN NOT IN USE, IDLE EQUIPMENT, PETROCHEMICALS, AND TOXIC/HAZARDOUS MATERIALS SHALL BE LOCKED AND MAY NOT BE STORED IN THE 100–YEAR FLOODPLAIN OR NEAR ANY DRAINAGE WAYS, DITCHES OR STREAMS; DISCHARGE OF PETROLEUM PRODUCTS, CEMENT WASHINGS, OR OTHER CONSTRUCTION MATERIALS INTO THE RIVER IS NOT PERMITTED.

19. ALL FUELING OPERATIONS, LUBRICATING, HYDRAULIC TOPPING OFF, FUEL TANK PURGING, AND EQUIPMENT MAINTENANCE/REPAIRS SHALL TAKE PLACE ON AN APPROVED PAD WITH SPILL CONTROL/COLLECTION DEVICES IN PLACE. THE USE OF CANOLA OIL OR OTHER BIODEGRADABLE FUELS AND FLUIDS IS RECOMMENDED WHEN WORKING IN SENSITIVE RIVERINE ENVIRONMENTS.

20. APPROPRIATE OIL SPILL KITS SHALL BE MAINTAINED ON SITE AND READILY ACCESSIBLE AT ALL TIMES DURING CONSTRUCTION AND EACH OPERATOR TRAINED IN ITS USE.

21. NO WASTEWATER SHALL BE DISCHARGED INTO THE RIVER.

22. PRIOR TO MOVING CONSTRUCTION EQUIPMENT INTO THE PROJECT AREA, THE CONTRACTOR MUST TAKE REASONABLE MEASURES TO ENSURE THAT EACH PIECE OF EQUIPMENT IS FREE OF SOIL, SEEDS, VEGETATIVE MATTER, OR OTHER DEBRIS THAT MAY CONTAIN SEEDS OF NON–NATIVE INVASIVE SPECIES (CONSIDER INCLUDING AQUATIC DECONTAMINATION MEASURES IF EQUIPMENT WILL BE USED IN–STREAM).

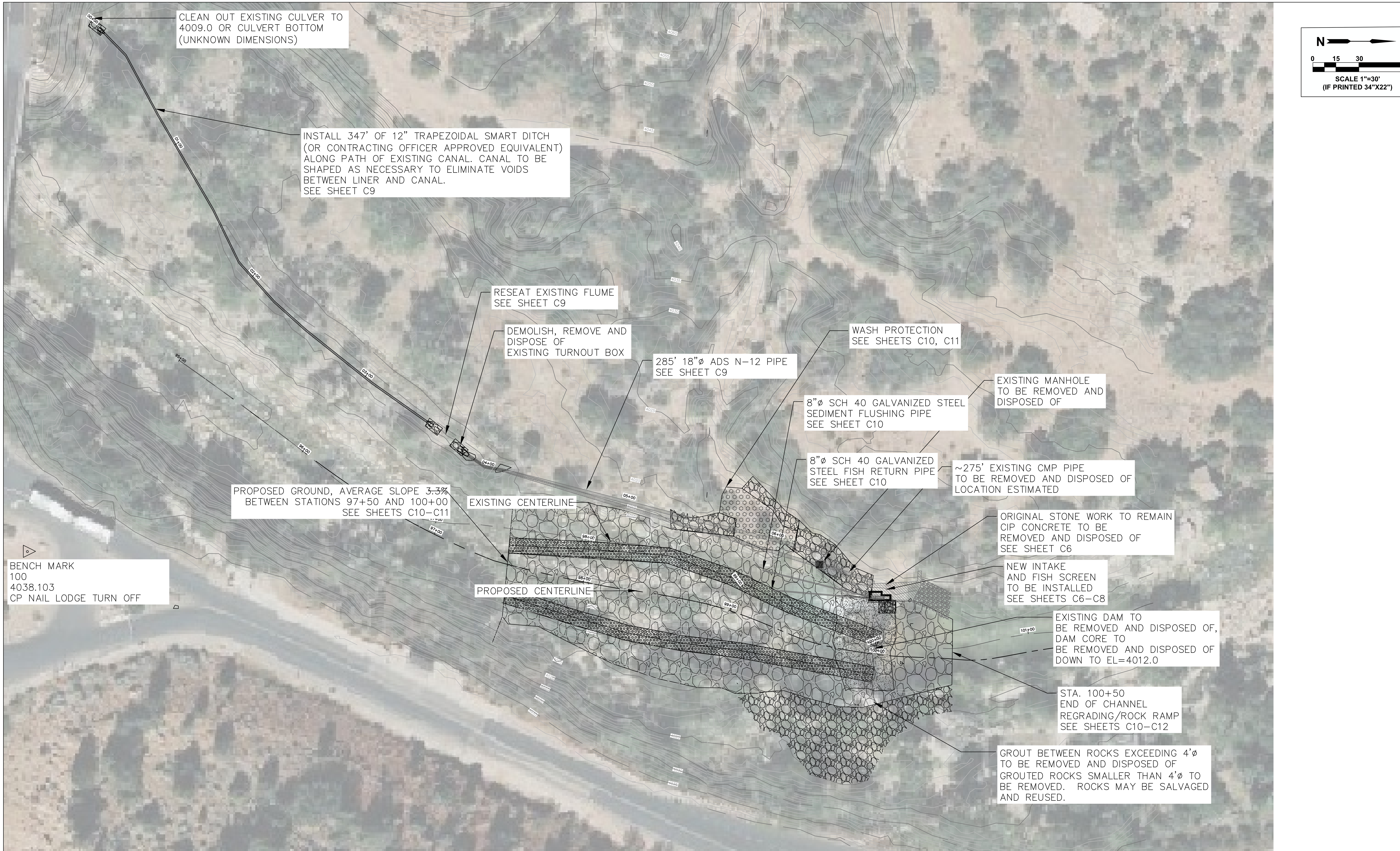
23. ALL TOOLS, EQUIPMENT, BARRICADES, SIGNS, SURPLUS MATERIALS, AND RUBBISH FROM THE PROJECT WORK LIMITS SHALL BE REMOVED UPON PROJECT COMPLETION. ALL CONSTRUCTION DEBRIS AND LITTER MUST BE COMPLETELY REMOVED OFF SITE AND DISPOSED OF PROPERLY UPON COMPLETION OF THE PROJECT.

24. ALL DEBRIS, EXCESS FILL MATERIAL AND EXCAVATED MATERIAL SHALL BE DISPOSED OF LEGALLY OFF–SITE BY THE CONTRACTOR.

25. STORMWATER POLLUTION PREVENTION PLAN (SWPPP) SHALL BE OBTAINED BY CONTRACTOR AND MAINTAINED DAILY.

PROFESSIONAL ENGINEER
2/1/26
11865813-2202
SUZANNE HUHTA
STATE OF UTAH

DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
SBH	C4	EROSION CONTROL NOTES	———
CADD: SBH			PMIS/PKG NO.
TECH. REVIEW: CAH		CRAWFORD DIVERSION FISH PASSAGE ZION NATIONAL PARK	SHEET
DATE: 02/01/26			6 OF 17

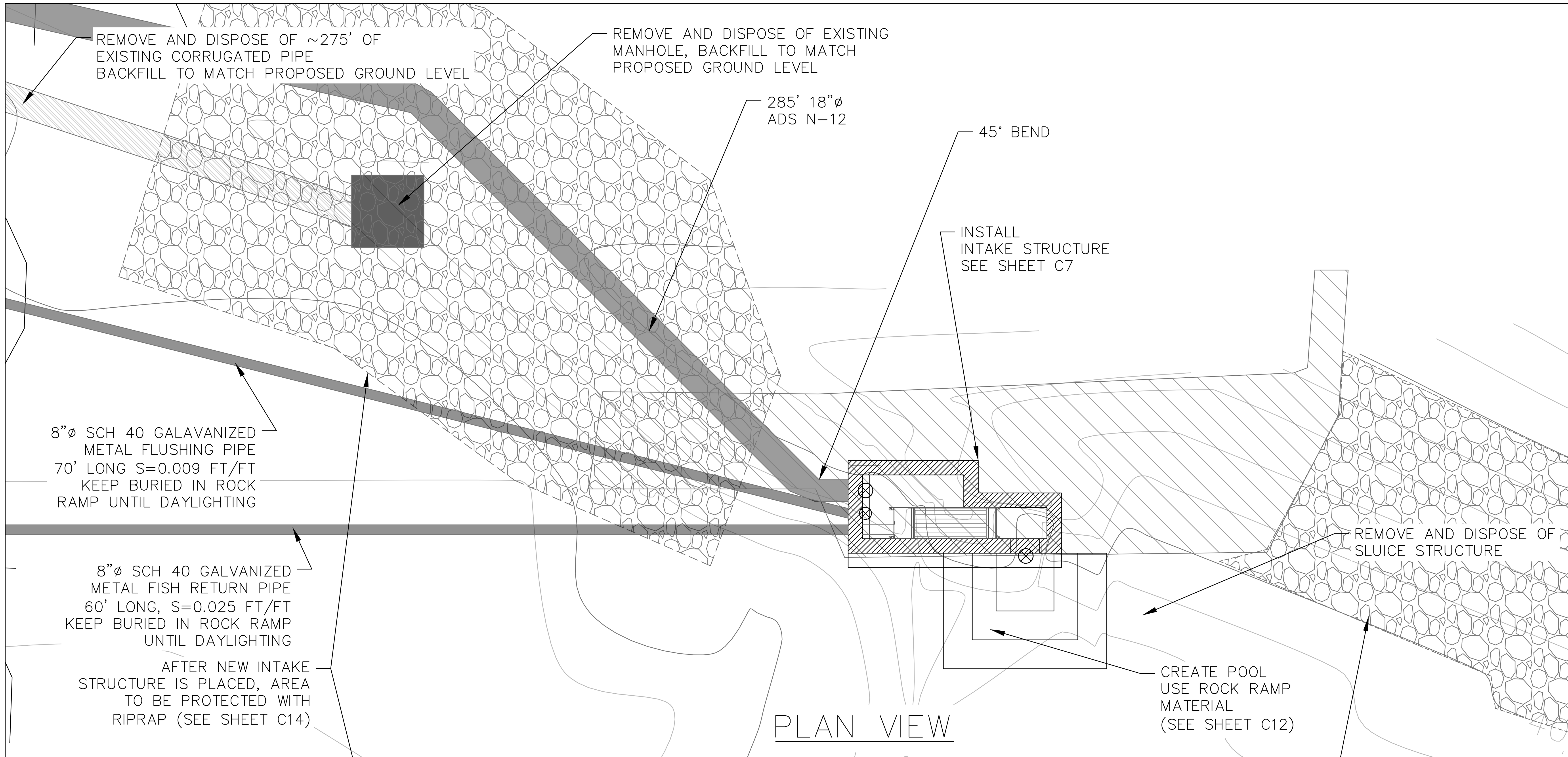


OVERALL SITE PLAN

PROFESSIONAL ENGINEER
2/1/26
11865813-2202
SUZANNE HUHTA
STATE OF UTAH

DESIGNED: SBH	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
CADD: SBH	C5	OVERALL SITE PLAN	——
TECH. REVIEW: CAH		CRAWFORD DIVERSION FISH PASSAGE ZION NATIONAL PARK	PMIS/PKG NO.
DATE: 02/01/26			SHEET 7 OF 17

FILE NAME: 2026-02 OVERALL C5.DWG
PLOT TIME/DATE: 3:58 PM, 4/9/2026
PLOTTED BY: Suzanne Huhta
PLOT STYLE: ----



PLAN VIEW

EXISTING INTAKE STRUCTURE

N

0 5 10 20

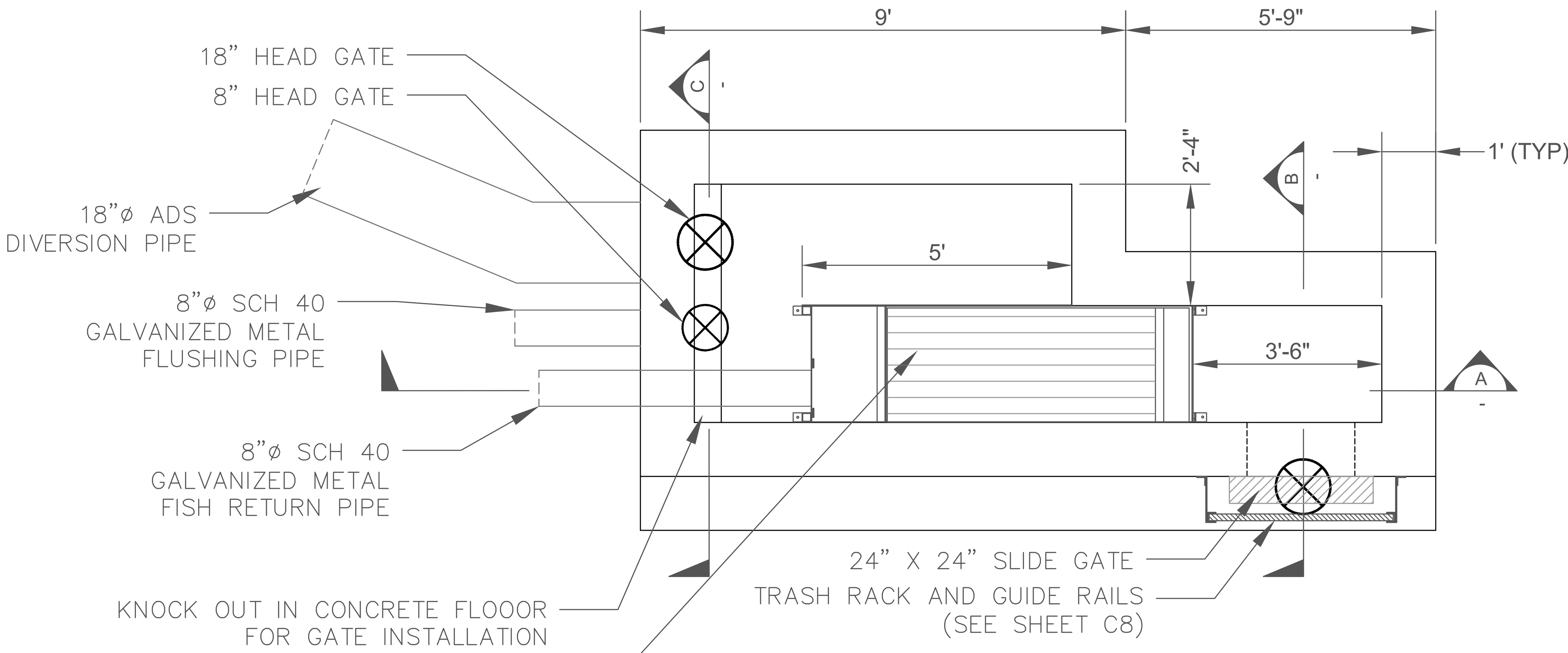
SCALE 1"=10'
(IF PRINTED 34" X 22")

NOTES

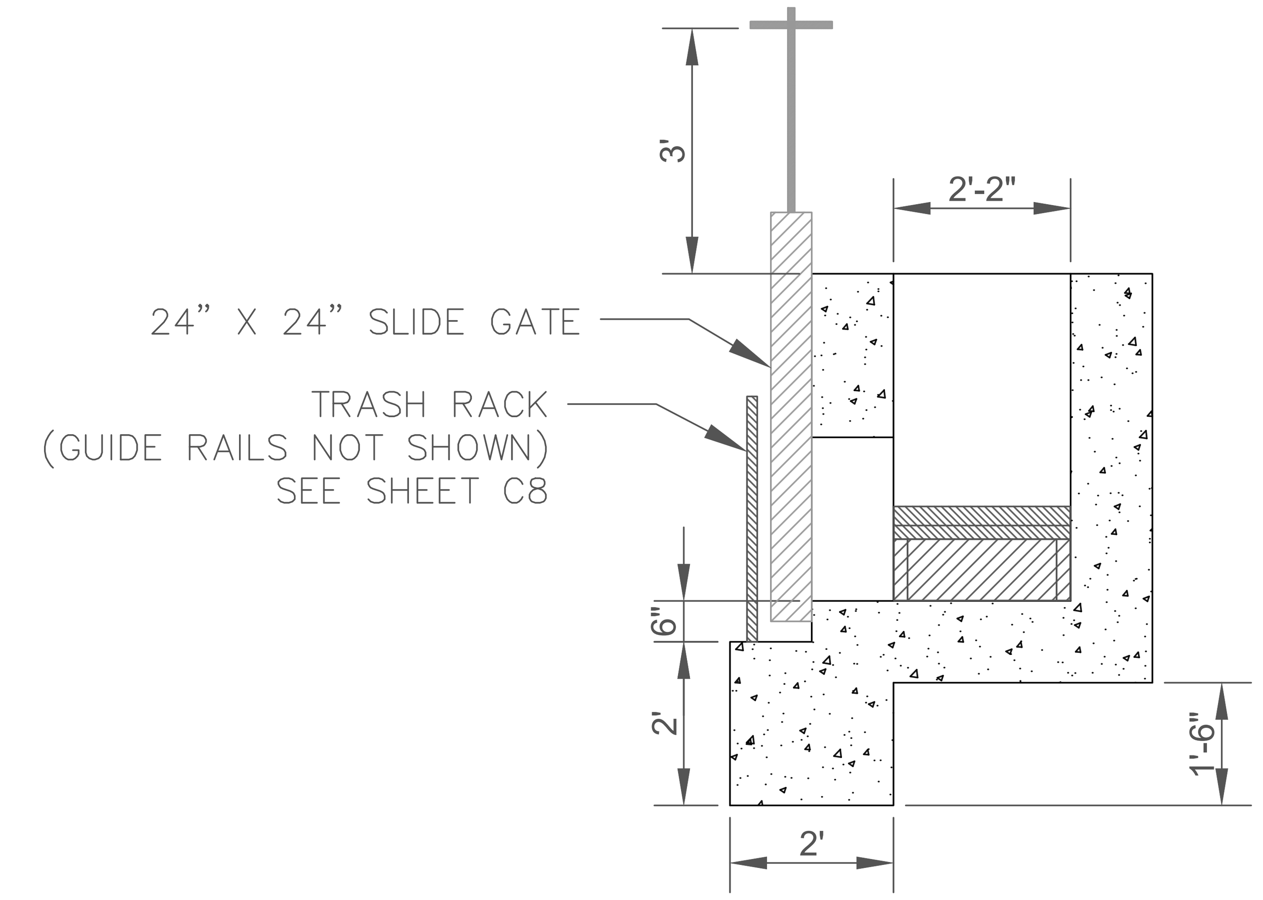
1. MISCELLANEOUS EXCESS CONCRETE FROM PREVIOUS WORK ON THE HEAD GATE STRUCTURE WAS POURED IN UPLAND AREA ABOVE THE STRUCTURE. THIS CONCRETE SHALL BE REMOVED AND DISPOSED OF AS PART OF THIS PROJECT.

PROFESSIONAL ENGINEER
2/1/26
11865813-2202
SUZANNE
HUHTA
STATE OF UTAH

DESIGNED: SBH CADD: SBH TECH. REVIEW: CAH DATE: 02/01/26	SUB SHEET NO. C6	TITLE OF SHEET INTAKE STRUCTURE AND UPPER WATERLINE PLAN CRAWFORD DIVERSION FISH PASSAGE ZION NATIONAL PARK	DRAWING NO. PMIS/PKG NO. SHEET 8 OF 17



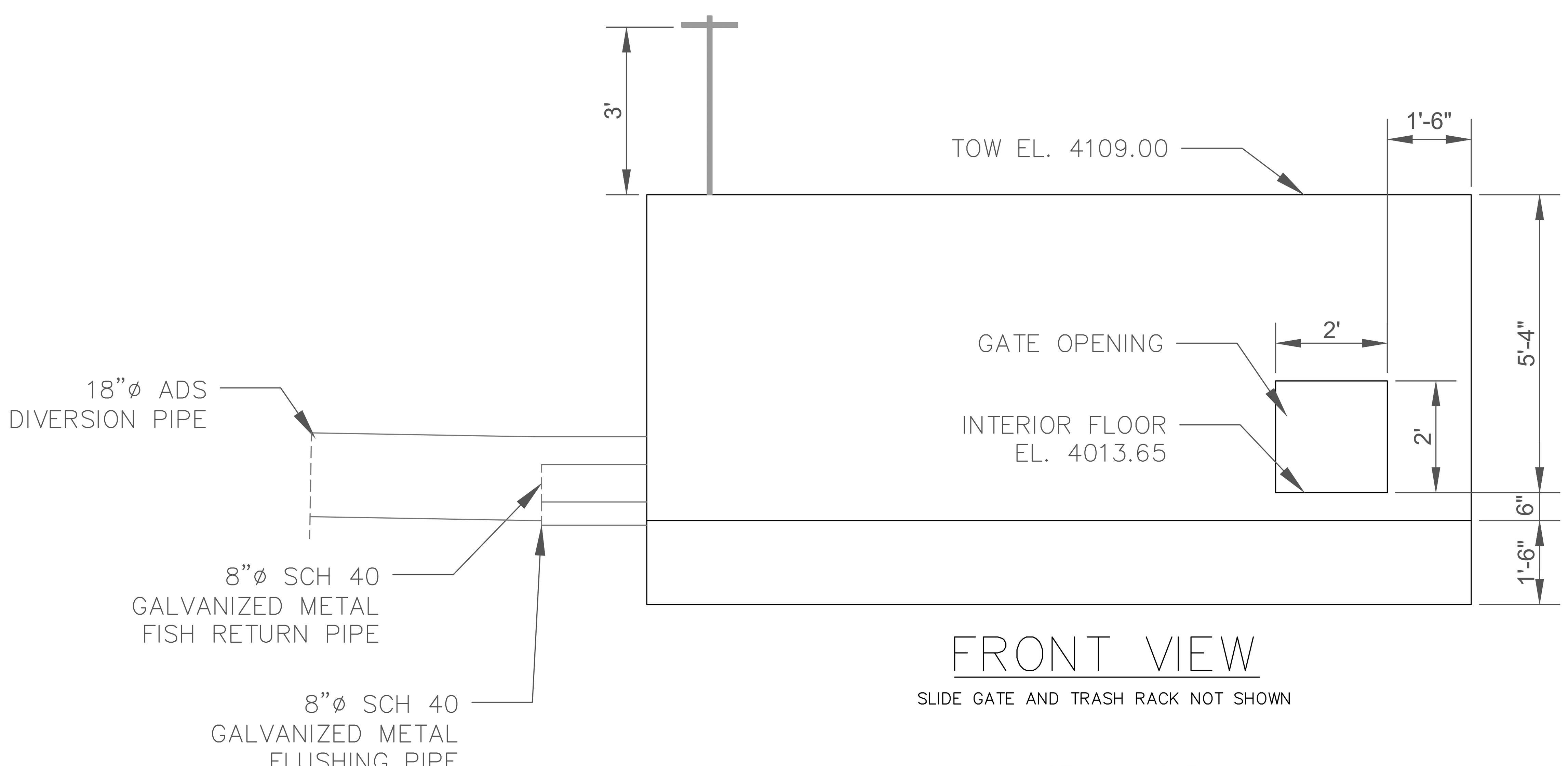
PLAN VIEW



SECTION B

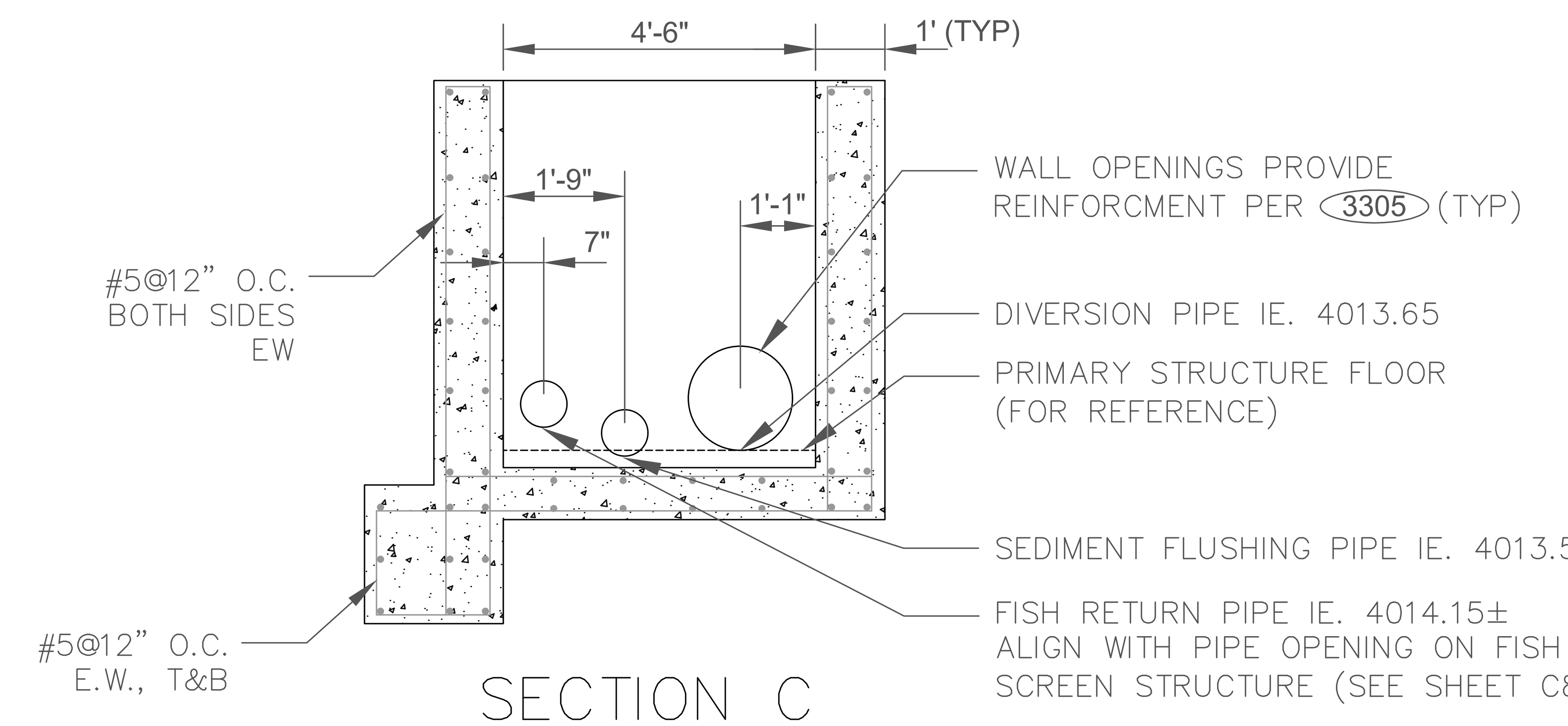
NOTES

1. 8" AND 18" HEAD GATES SHALL BE WATERMAN C–10, FRESNO SERIES 6600 OR EQUIVALENT APPROVED BY C.O. FRAME MATERIAL SHALL BE GALVANIZED STEEL.
2. 24" X 24" SLIDE GATE SHALL BE FRESNO SERIES 8200 OR EQUIVALENT APPROVED BY C.O. FRAME MATERIAL SHALL BE GALVANIZED STEEL.
3. GATE INSTALLATION SHALL FOLLOW MANUFACTURER RECOMMENDATIONS AND BE APPROVED BY C.O.
4. CONTRACTOR TO PROVIDE DETAILS FOR ALL PIPE TO WALL/STRUCTURE CONNECTIONS AND LOCATIONS FOR C.O. APPROVAL.
   4.1. ONE OPTION FOR CONNECTION OF ADS PIPE TO CONCRETE WALLS IS DUAL WALL ADAPTER FITTING BY ADS, P/N 1821AN.
5. HYDROPHILIC WATERSTOP SHALL BE INSTALLED AROUND COMPLETE CIRCUMFERENCE OF ALL PIPE/WALL CONNECTIONS.



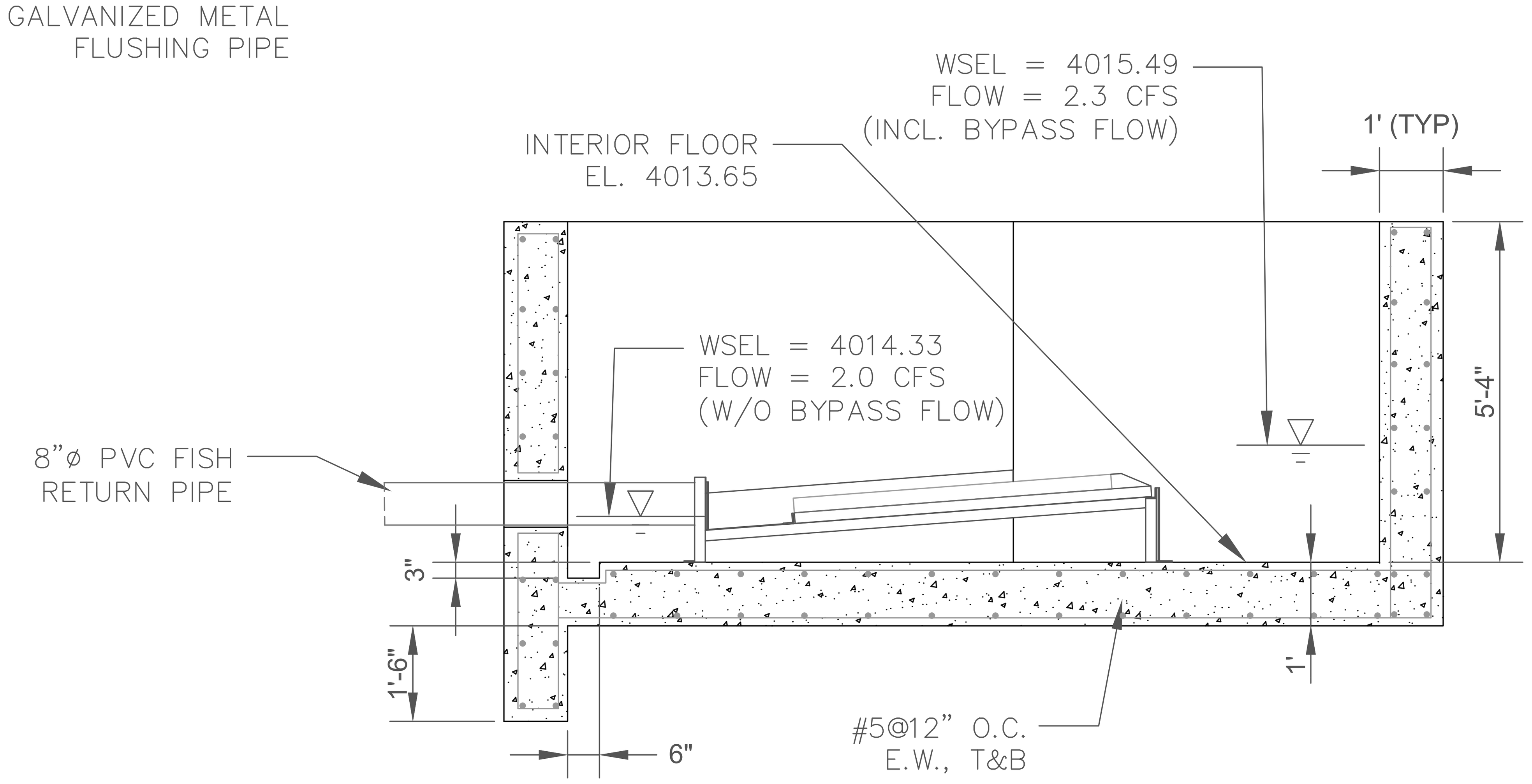
FRONT VIEW

SLIDE GATE AND TRASH RACK NOT SHOWN



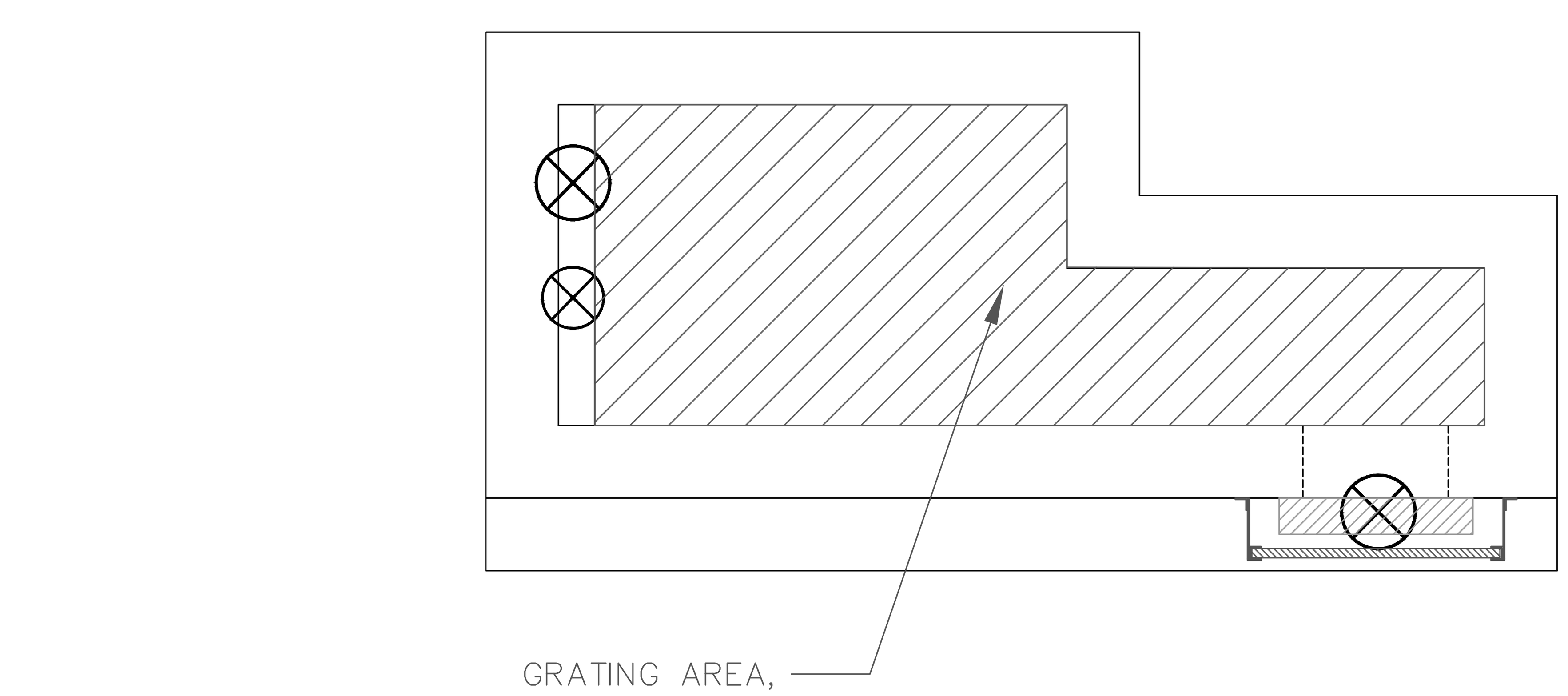
SECTION C

GATES NOT SHOWN

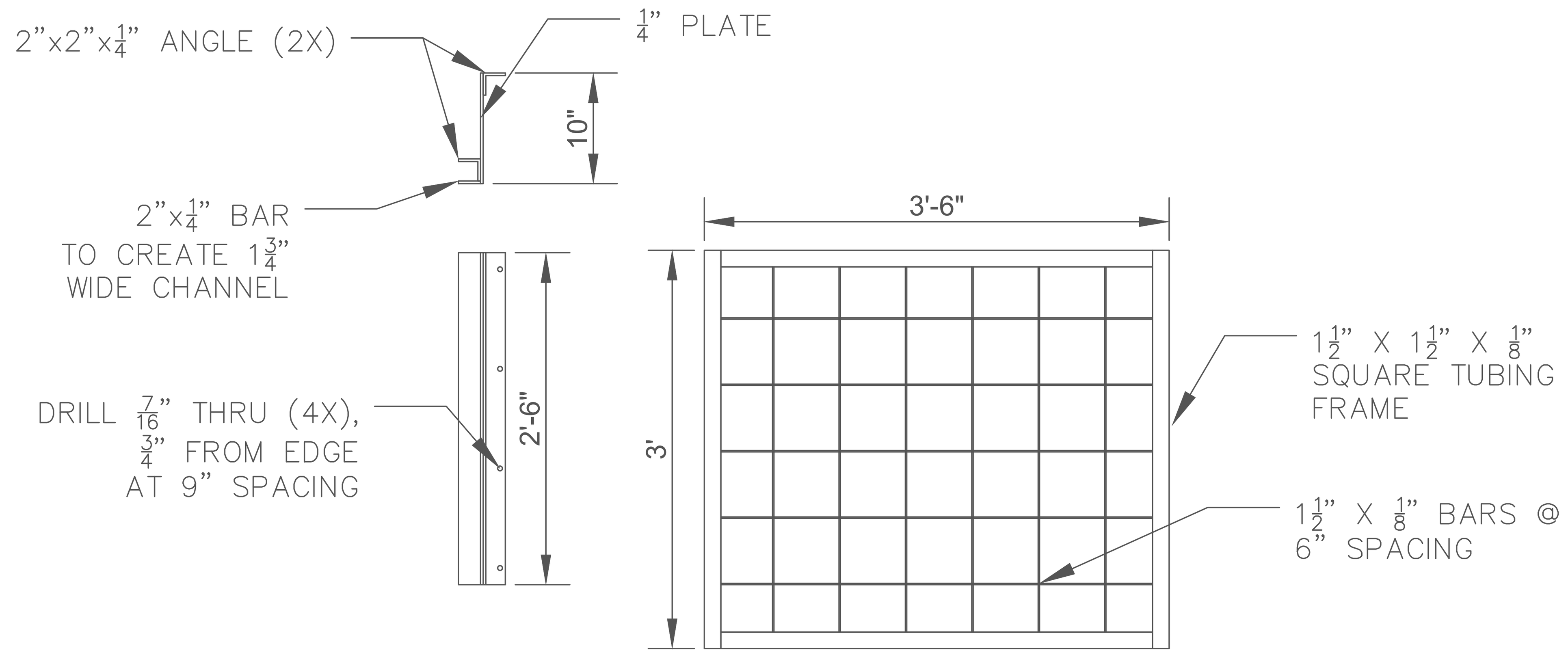


SECTION A

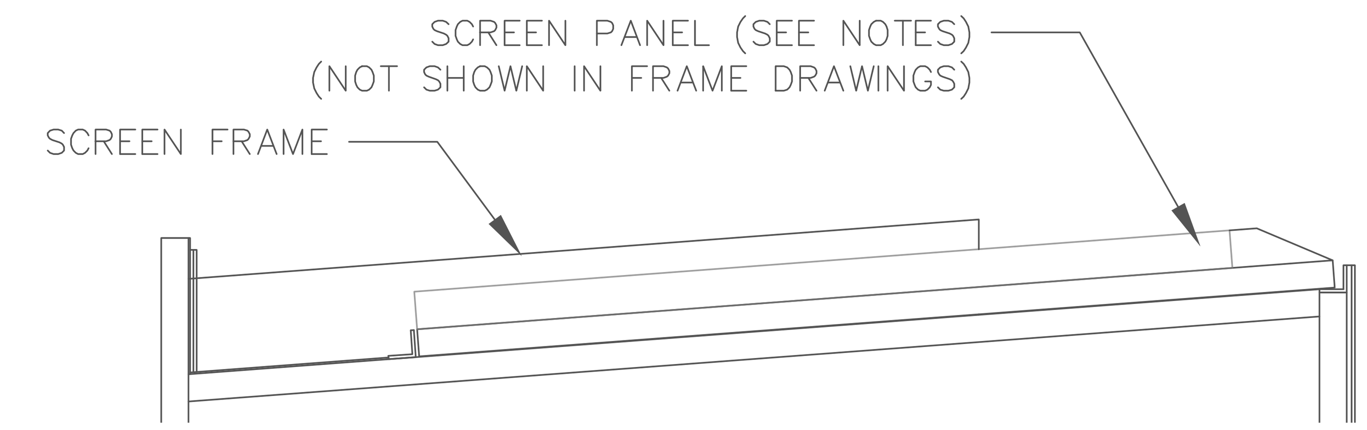
GATES AND DIVERSION/SEDIMENT PIPES NOT SHOWN



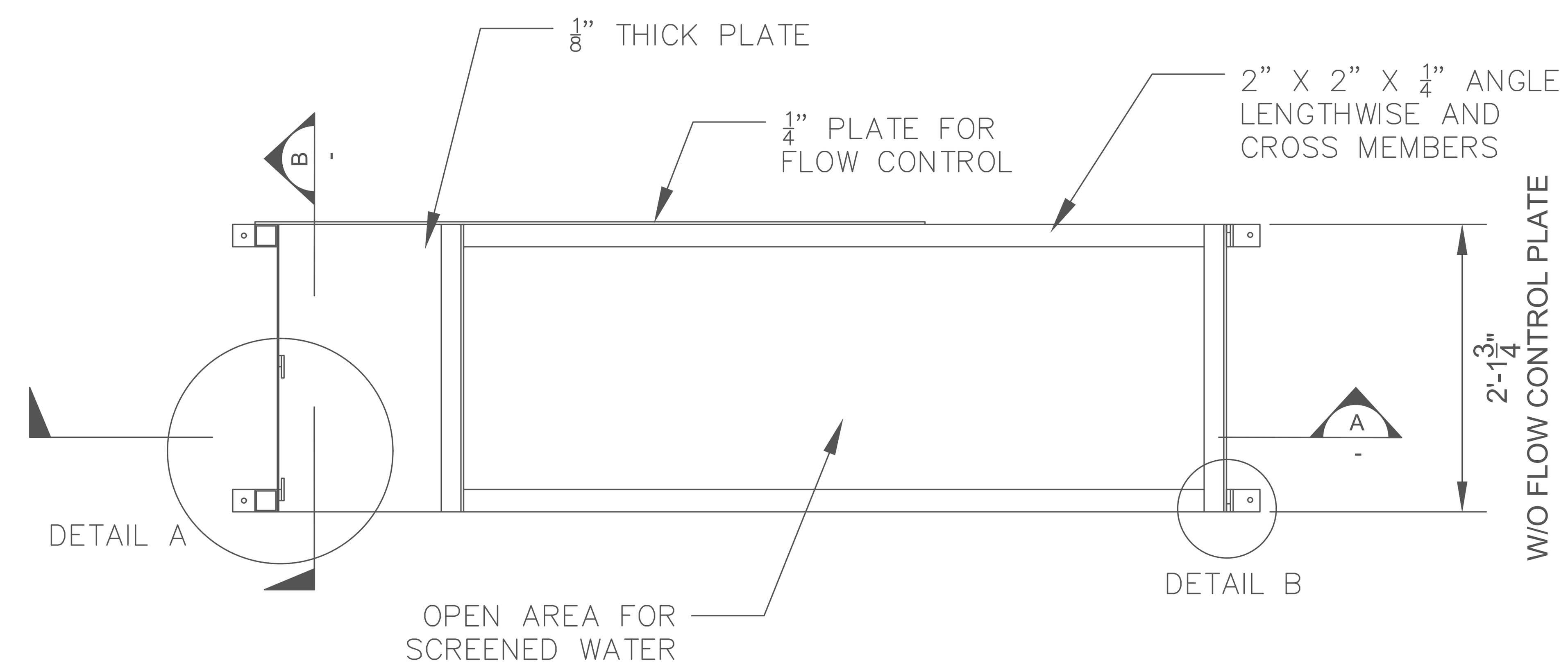
DESIGNED: SBH	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
CADD: SBH	C7	INTAKE STRUCTURE	———
TECH. REVIEW: CAH			PMIS/PKG NO.
DATE: 02/01/26		CRAWFORD DIVERSION FISH PASSAGE ZION NATIONAL PARK	SHEET 9 OF 17



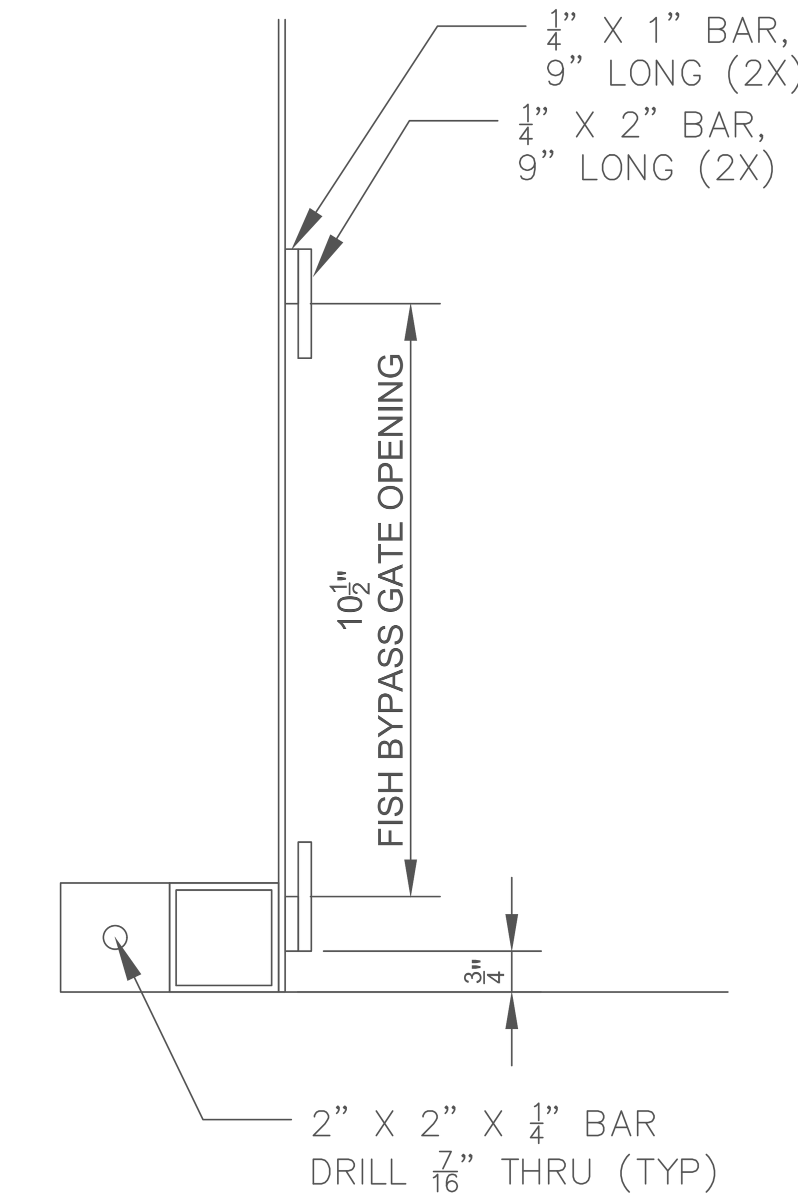
GUIDE RAILS (2X) AND TRASH RACK (1X)



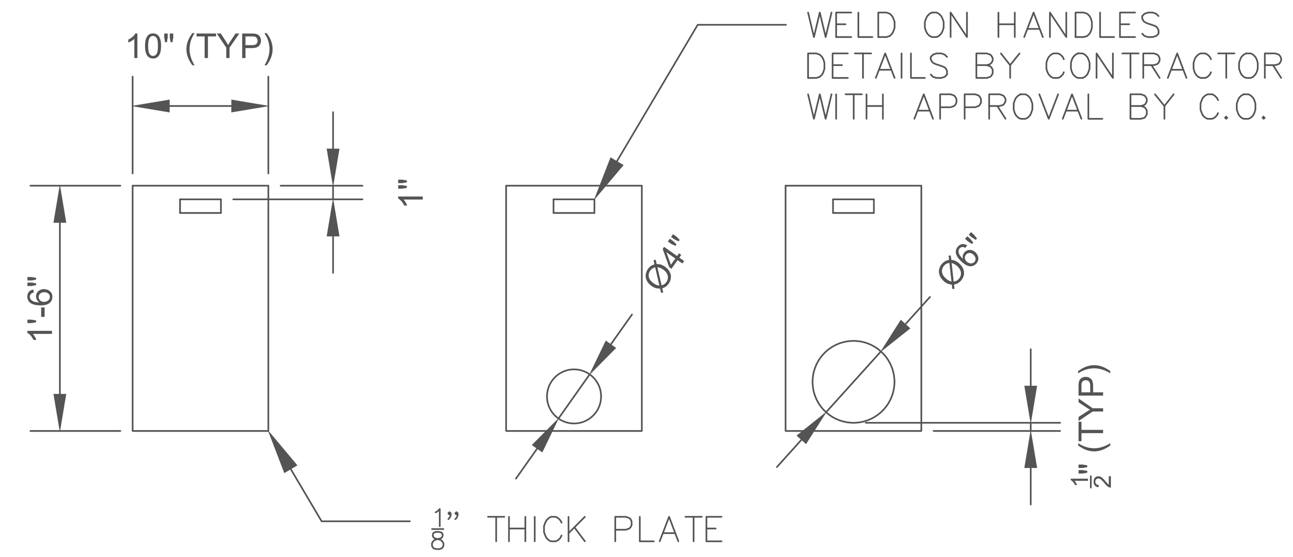
SCREEN OVERVIEW (ELEVATION VIEW)



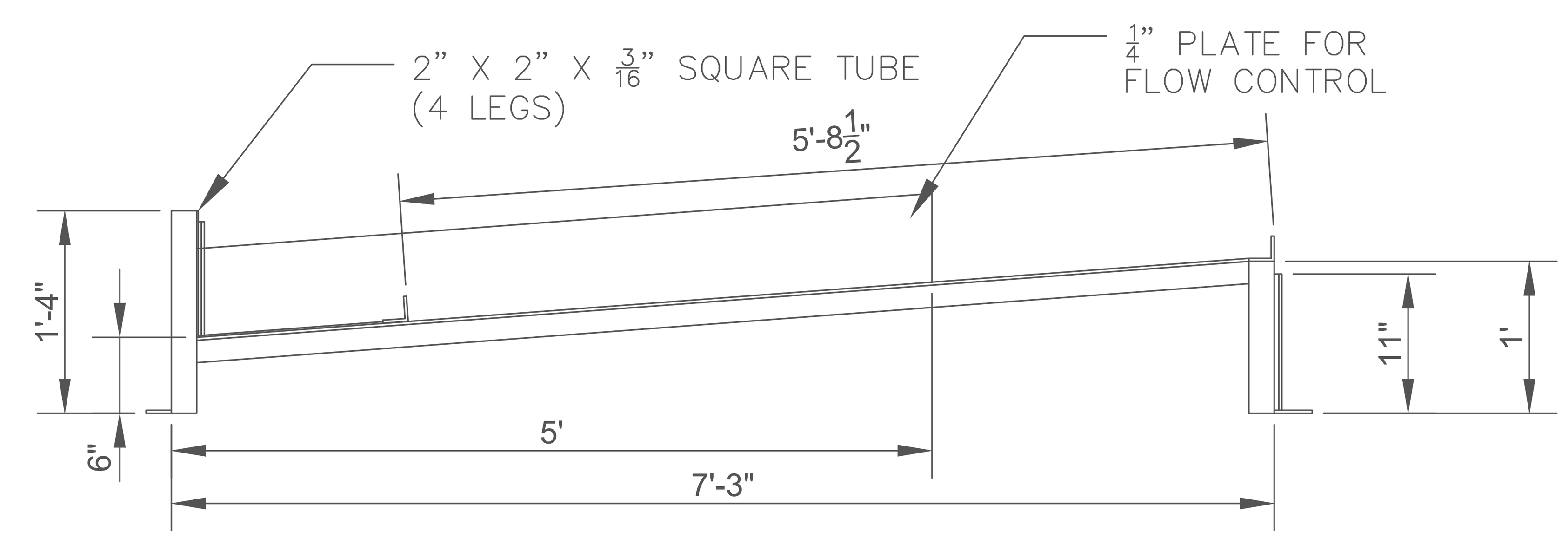
PLAN VIEW



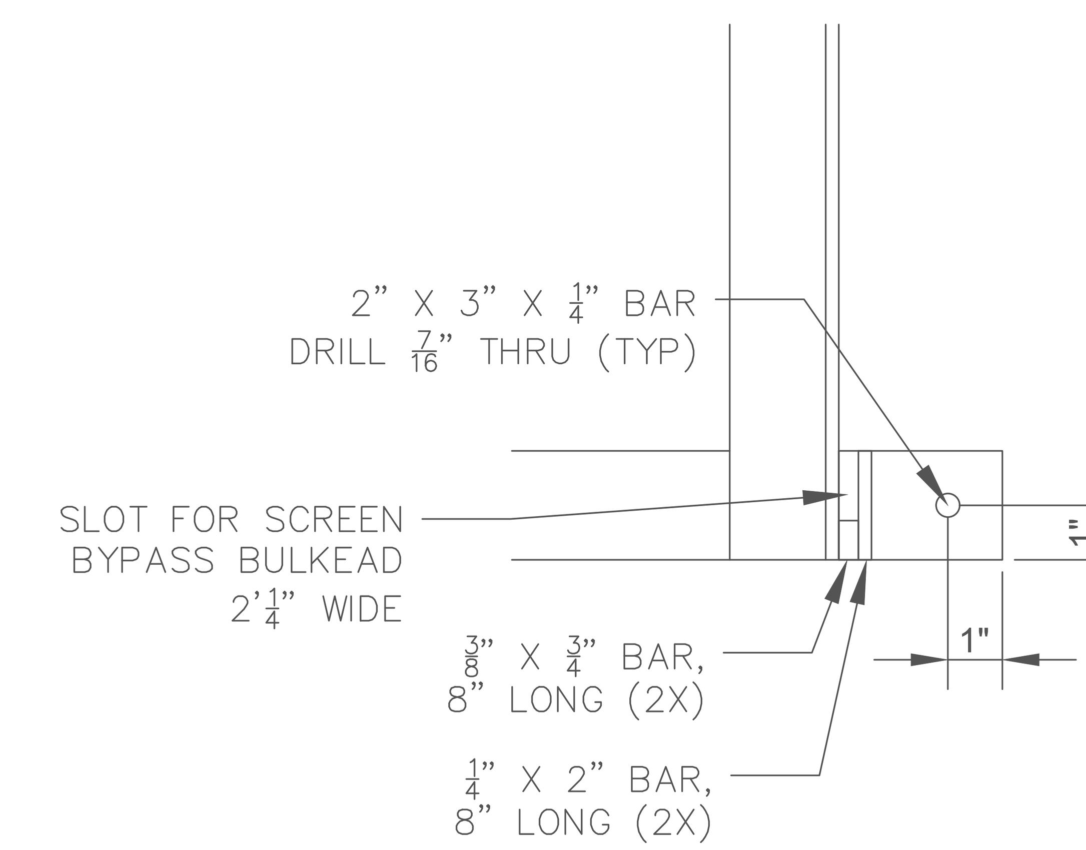
DETAIL A



FISH BYPASS GATES (3X)

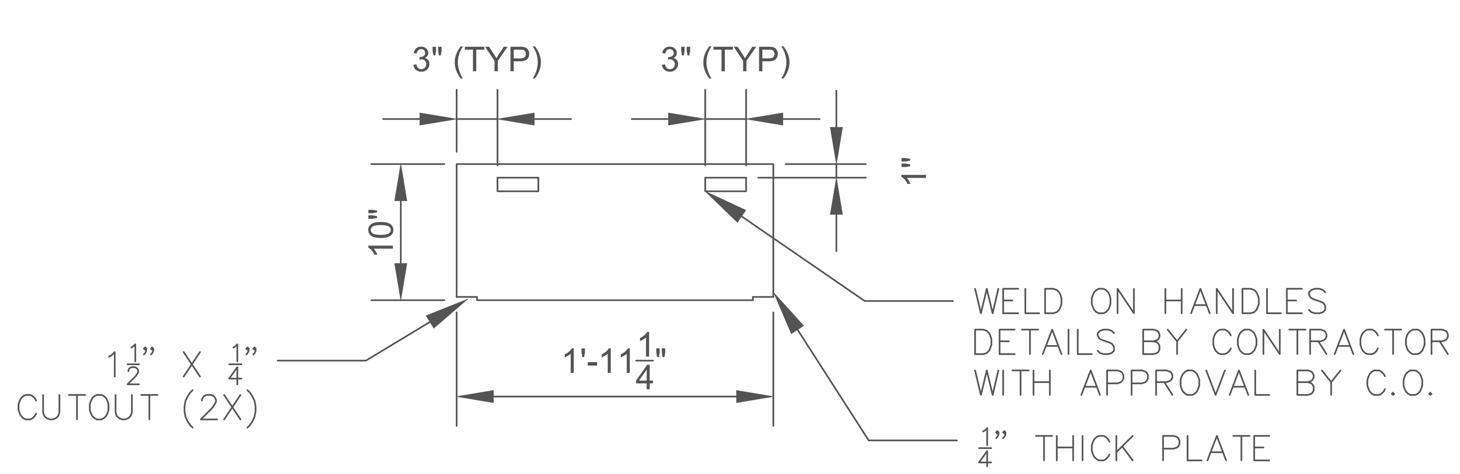


SECTION A

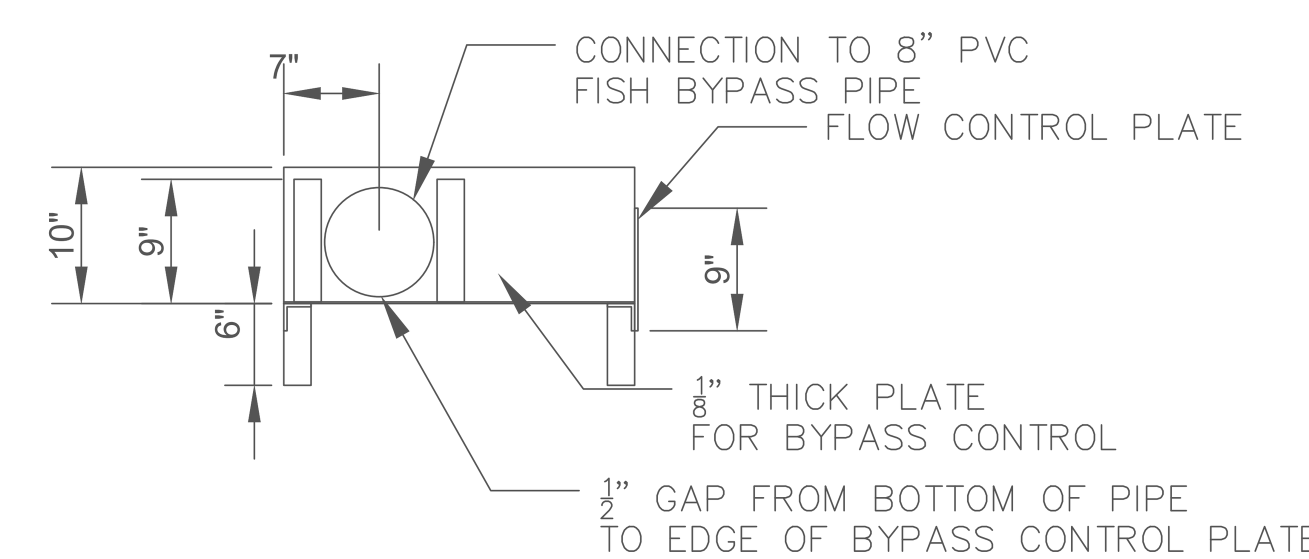


DETAIL B

TYP EACH SIDE



SCREEN BYPASS BULKHEAD

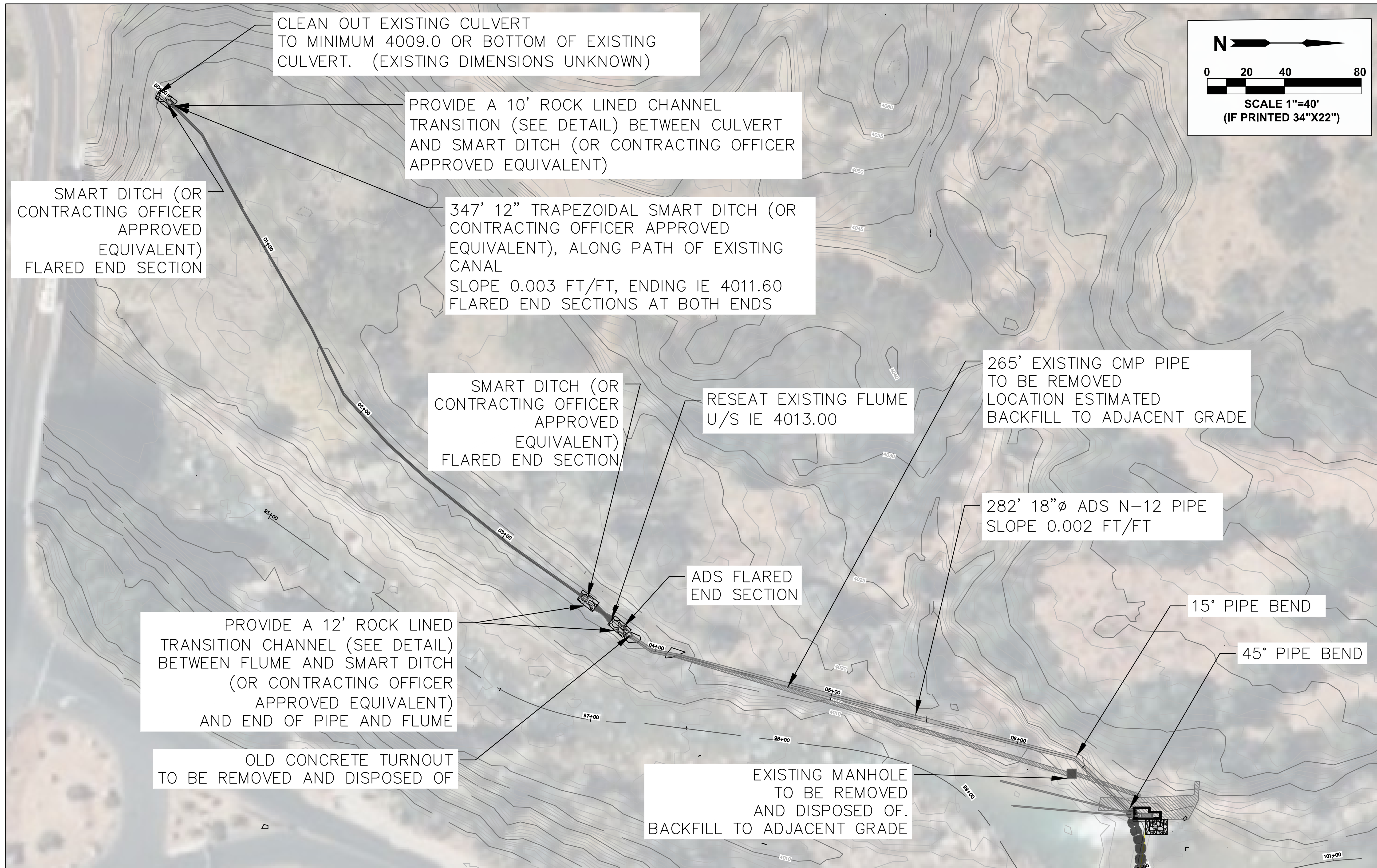


SECTION B

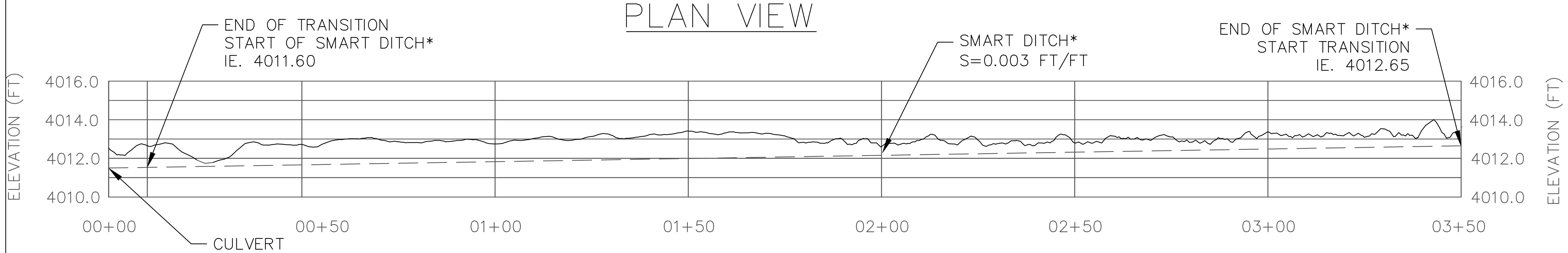
NOTES

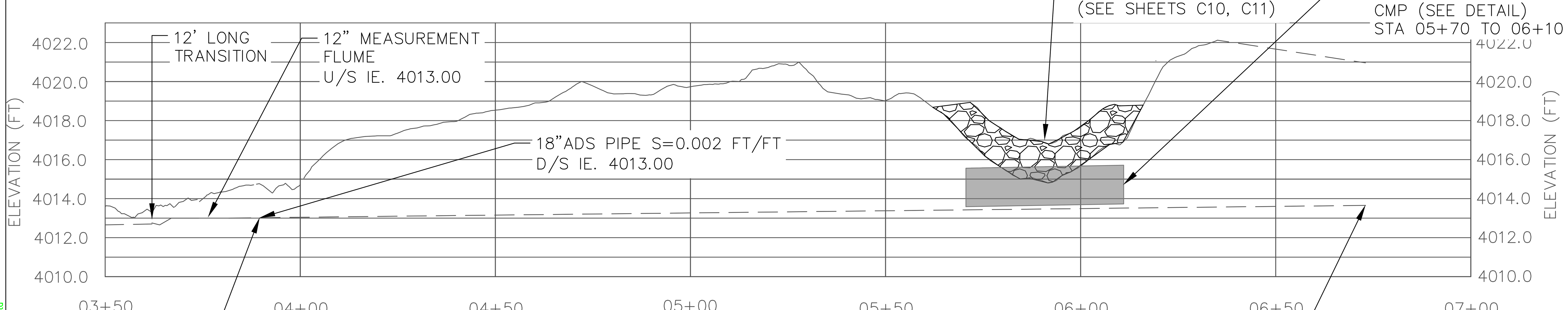
1. ALL METAL STRUCTURES SHALL BE HOT DIP GALVANIZED STEEL.
2. METAL SEAMS TO BE FILLET WELDED BOTH SIDES BASED ON SMALLER PLATE THICKNESS: 1/8" WELD FOR 1/8" PLATE, 3/16" WELD FOR 1/4" PLATE, 1/4" WELD FOR 3/8" PLATE.
3. TRASH RACK GUIDE RAILS TO BE SECURED TO WALL USING 3/8" SS CONCRETE ANCHORS (8X)
4. SCREEN PANEL MANUFACTURED BY CORRUGATED WATER SCREENS. DIMENSIONS 25 3/8" W X 67 3/8" L X 4 13/16" H, 1/8" HOLES (42% OPEN AREA). PANEL RESTS ON SCREEN FRAME BUT IS NOT PERMANENTLY ATTACHED. SCREEN PANEL SHALL INCLUDE A MINIMUM OF TWO (2) BRUSHES DESIGNED FOR CLEANING THE SCREEN PANEL (PROVIDED BY CORRUGATED WATER SCREENS).
5. SCREEN FRAME AND TRASH RACK GUIDE RAILS TO BE SECURED TO INTAKE STRUCTURE USING 3/8" GALVANIZED CONCRETE ANCHORS (4X) AT THE LOCATION SHOWN ON SHEET C7.
6. CONTRACTOR TO FILL AND SEAL ALL GAPS BETWEEN SCREEN FRAME, SCREEN PANEL, AND INTAKE STRUCTURE WALLS SUBJECT TO C.O. APPROVAL.

DESIGNED: SBH	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
CADD SBH	C8	FISH SCREEN DETAILS	
TECH. REVIEW: CAH			PMIS/PKG NO.
DATE: 02/01/26		CRAWFORD DIVERSION FISH PASSAGE ZION NATIONAL PARK	SHEET 10 OF 17



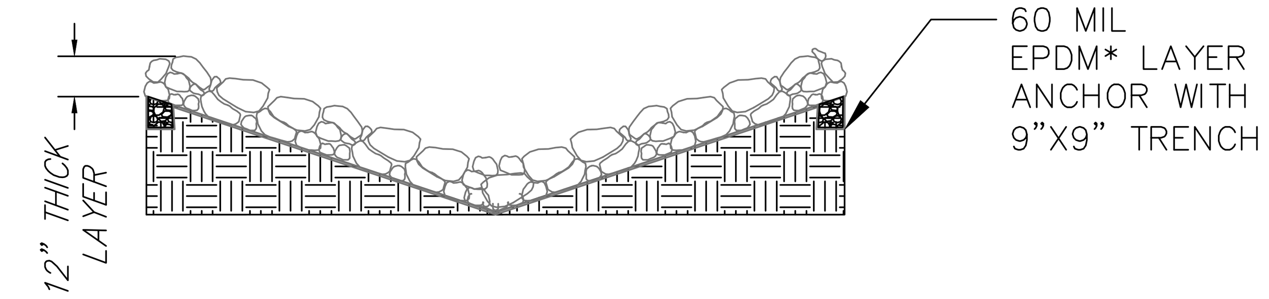
PLAN VIEW





SMART DITCH* AND PIPE PROFILE

*OR CONTRACTING OFFICER APPROVED EQUIVALENT



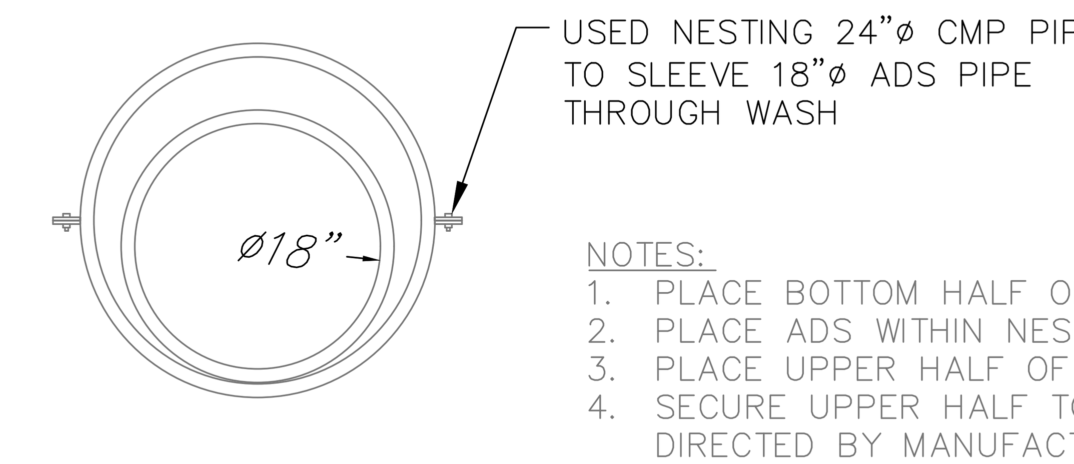
*EPDM – ETHYLENE PROPYLENE DIENE MONOMER

TRANSITION GRADATION	
% SMALLER THAN GIVEN SIZE BY WEIGHT	INTERMEDIATE ROCK DIMENSION (INCHES)
D_{15}	4
D_{50}	6
D_{85}	8.5
D_{100}	12

TRANSITION NOTES:
1. CHANNEL SLOPE TO MATCH UPSTREAM AND DOWNSTREAM SLOPES.
2. CHANNEL SIDE SLOPES TO MATCH UPSTREAM AND DOWNSTREAM CROSS SECTIONS, CREATING A SMOOTH TRANSITION BETWEEN THE UPSTREAM AND DOWNSTREAM CROSS SECTIONS.
3. GENERAL PLACEMENT TECHNIQUES SHOULD RESULT IN LARGER ROCK AT THE SURFACE AND ROCK SECURELY INTERLOCKED AT THE DESIGN THICKNESS AND GRADE. COMPACTION AND LEVELING SHOULD RESULT IN MINIMAL VOIDS AND PROJECTIONS ABOVE GRADE.

TRANSITION CHANNEL DETAILS

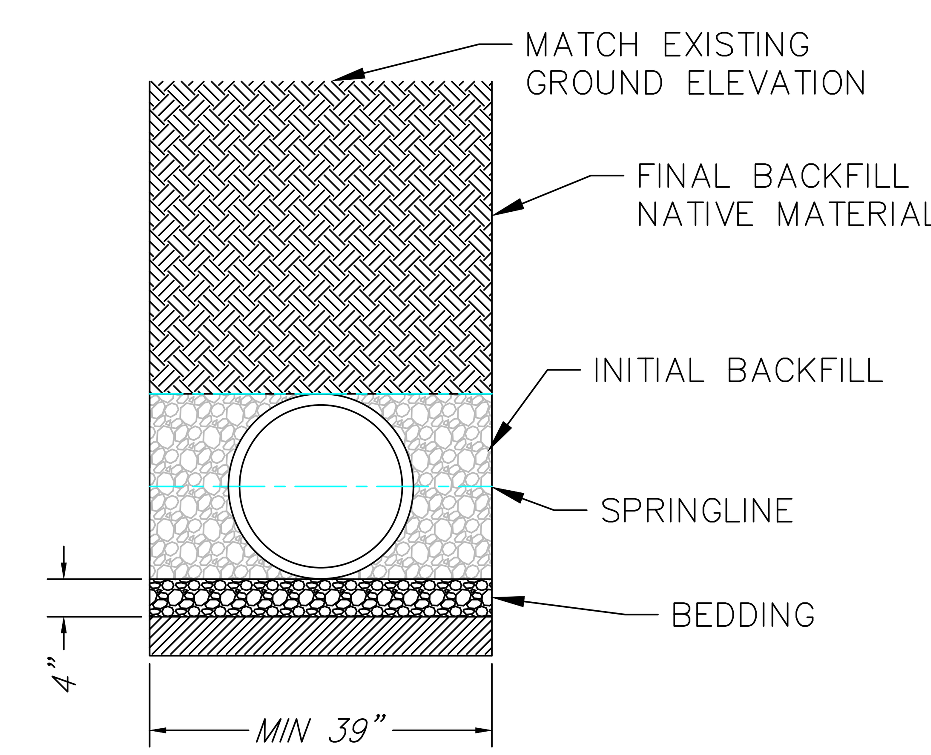
(TRANSITIONS ARE LOCATED BETWEEN PIPE AND FLUME, FLUME AND SMART DITCH*, AND SMART DITCH* AND CULVERT)
*OR CONTRACTING OFFICER APPROVED EQUIVALENT



NOTES:
1. PLACE BOTTOM HALF OF NESTING CMP PIPE
2. PLACE ADS WITHIN NESTING CMP PIPE
3. PLACE UPPER HALF OF NESTING CMP
4. SECURE UPPER HALF TO LOWER HALF AS DIRECTED BY MANUFACTURER.

PROTECTED ADS PIPE THROUGH WASH DETAIL

STA 05+70 TO 06+10



NOTES:
1. ALL PIPE SHALL BE INSTALLED IN ACCORDANCE WITH ASTM D2321, "STANDARD PRACTICE FOR UNDERGROUND INSTALLATION OF THERMOPLASTIC PIPE FOR SEWERS AND OTHER GRAVITY FLOW APPLICATIONS", LATEST EDITION.
2. ADS N-12 PIPE TO BE SOIL TIGHT BELL AND SPIGOT.
3. BEDDING AND INITIAL BACKFILL SHALL BE CRUSHED ANGULAR ROCK 1.5" OR LESS.

PROTECTED ADS PIPE THROUGH WASH DETAIL

PROFESSIONAL ENGINEER
11865813-2202
SUZANNE HUHTA
STATE OF UTAH

DESIGNED: SBH	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
CADD: SBH	C9	IRRIGATION CANAL AND PIPE PLAN, PROFILE AND DETAILS	——
TECH. REVIEW: CAH		CRAWFORD DIVERSION FISH PASSAGE	PMIS/PKG NO.
DATE: 02/01/26		ZION NATIONAL PARK	SHEET 11 OF 17

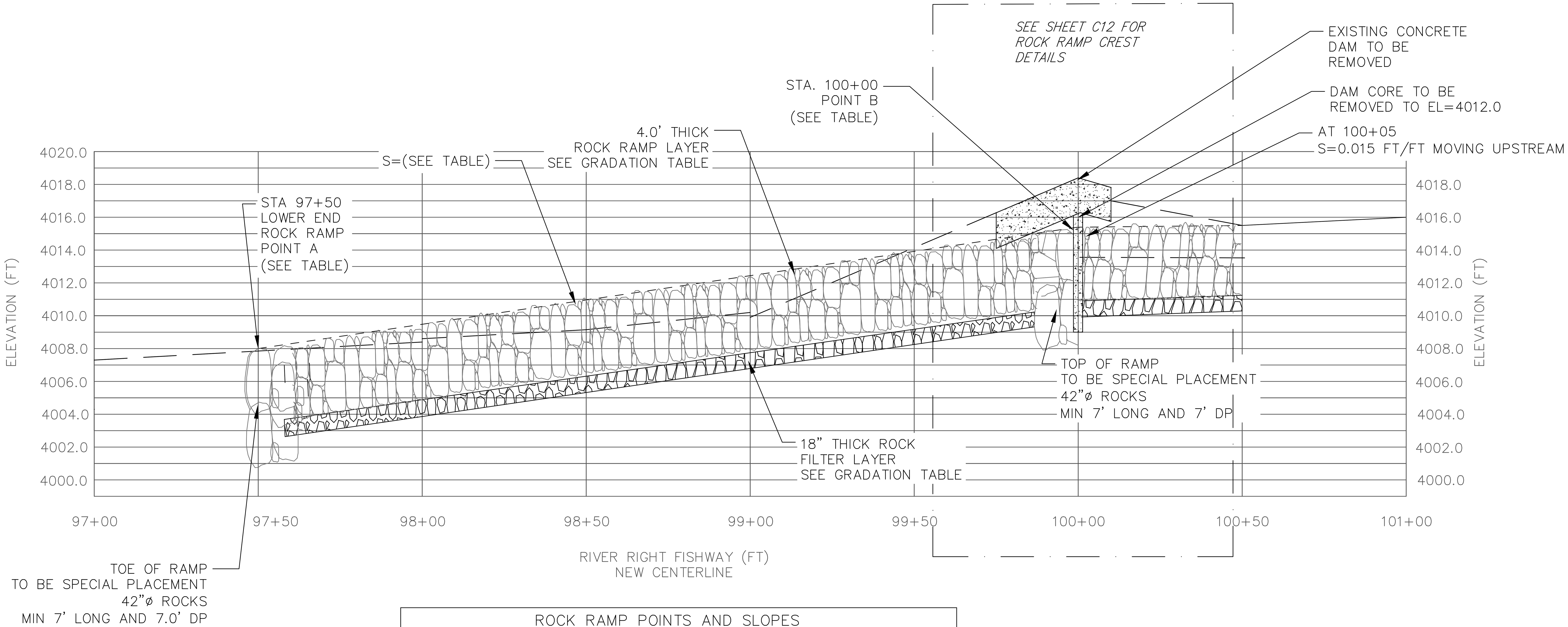
FILE NAME: 2026-02 CRAWFORD SMART DITCH & DIVERSION PIPE C9.DWG
PLOT TIME/DATE: 3:33 PM, 4/8/2026
PLOTTED BY: Suzanne Huhta
PLOT STYLE: -----

RIVER PLAN VIEW

PROFESSIONAL ENGINEER
11865813-2202
SUZANNE HUHTA
STATE OF UTAH

DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
SBH	C10	RIVER AND WASH PLAN	
CADD: SBH			PMIS/PKG NO.
TECH. REVIEW: CAH		CRAWFORD DIVERSION FISH PASSAGE	SHEET
DATE: 02/01/26		ZION NATIONAL PARK	12 OF 17

FILE NAME: 2026-20 RIVER PLAN AND PROFILE C10C11.DWG
PLOT TIME/DATE: 1:29 PM, 2/1/2026
PLOTTED BY: Suzanne Huhta
PLOT STYLE: Acad.ctb

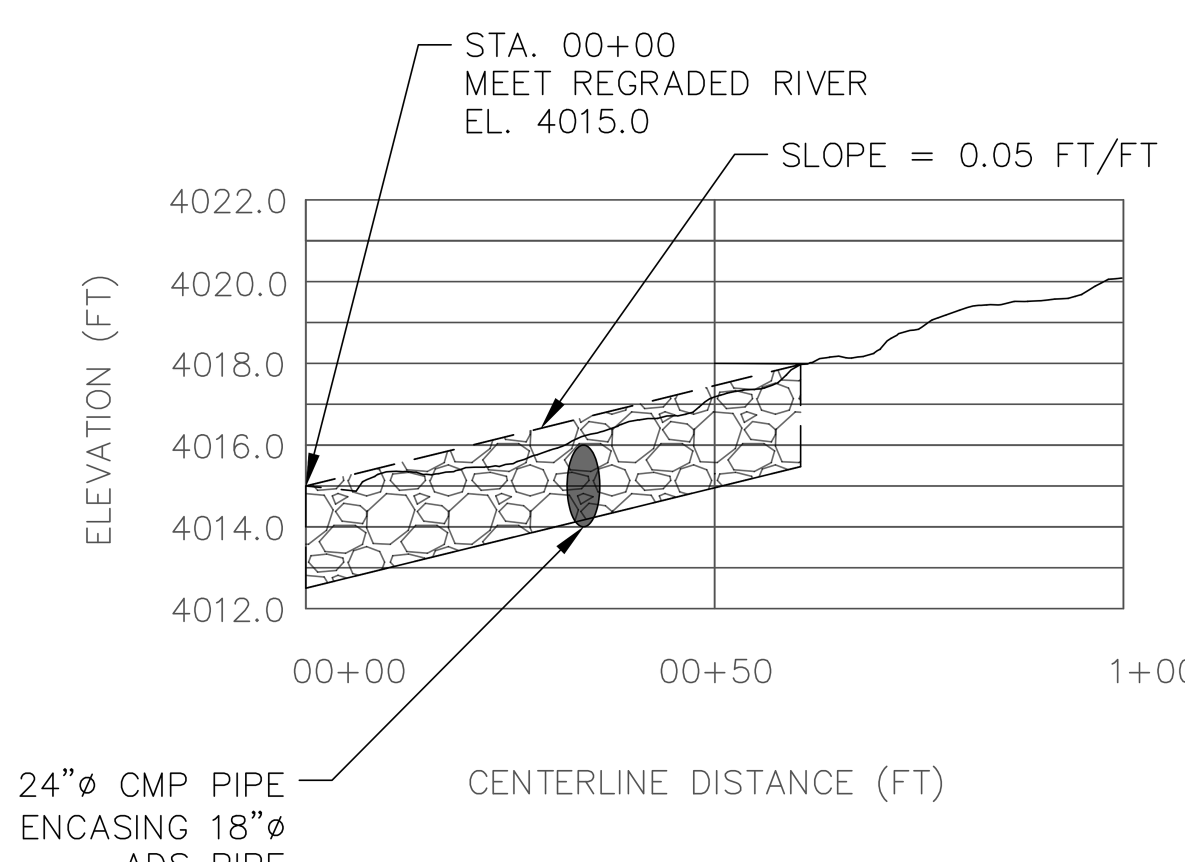


ROCK RAMP POINTS AND SLOPES			
	SLOPE	POINT A STA. 97+50	POINT B STA. 100+00
RIVER RIGHT FISHWAY	0.029 FT/FT	EL. 4008.00	EL. 4015.25
ALONG CENTERLINE OF ROCK RAMP	0.033 FT/FT	EL. 4008.00	EL. 4016.25
RIVER LEFT FISHWAY	0.028 FT/FT	EL. 4008.50	EL. 4015.5

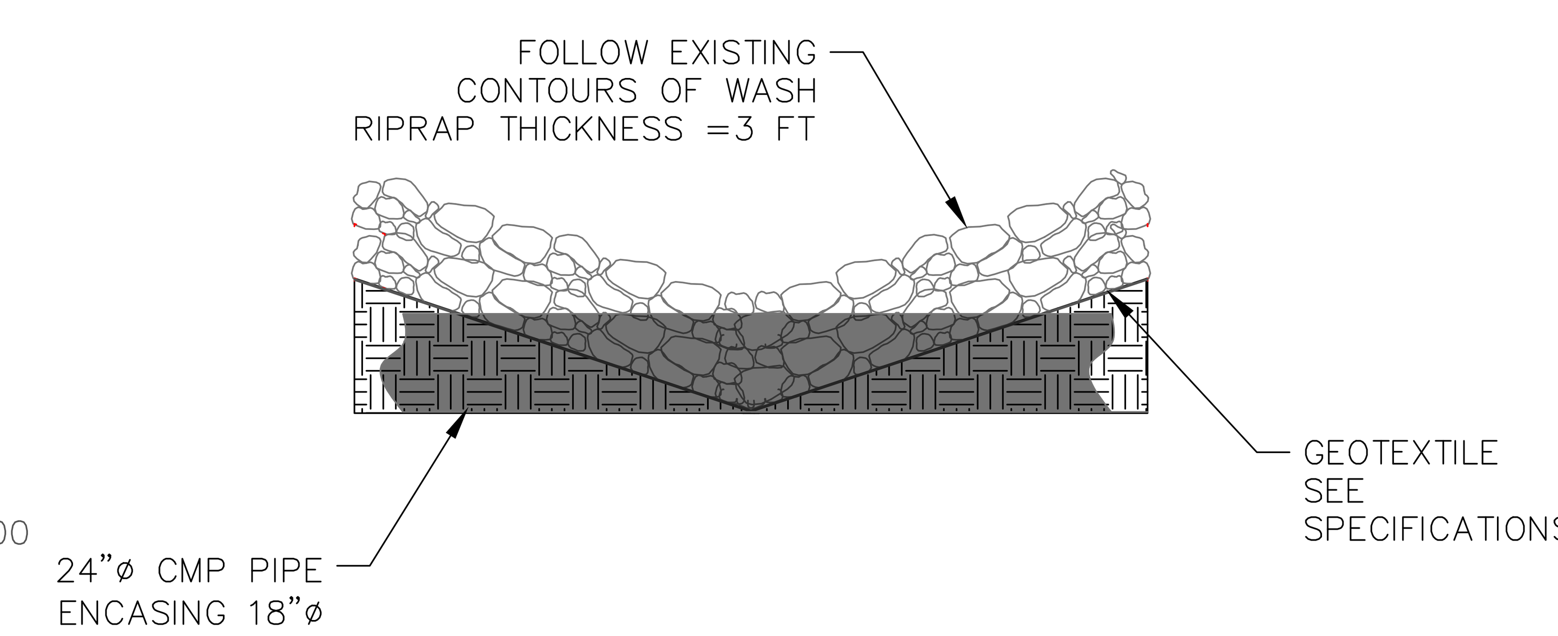
ROCK RAMP PROFILE

4.0' THICK BED MATERIAL GRADATION	
% SMALLER THAN GIVEN SIZE BY WEIGHT	INTERMEDIATE ROCK DIMENSION (INCHES)
D_{15}	15
D_{50}	24
D_{85}	37
D_{100}	48
NATIVE FINES <0.5" TO BE PRESSURE WASHED INTO VOIDS. SEE SHEET C15	

18" THICK FILTER MATERIAL GRADATION	
% SMALLER THAN GIVEN SIZE BY WEIGHT	INTERMEDIATE ROCK DIMENSION (INCHES)
D_{15}	0.5
D_{50}	3.0
D_{85}	4.5
D_{100}	6.0



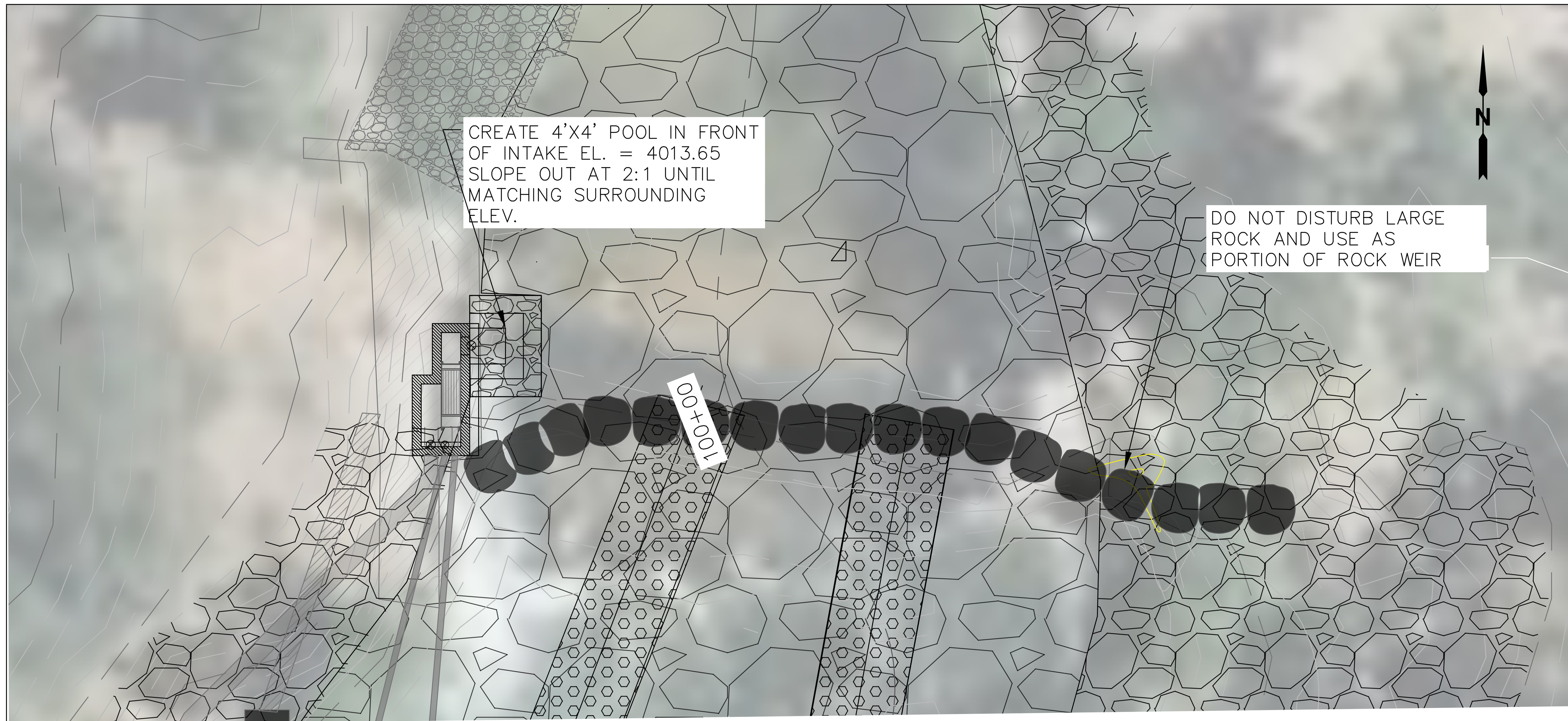
WASH PROFILE



WASH CROSS SECTION

WASH GRADATION	
% SMALLER THAN GIVEN SIZE BY WEIGHT	INTERMEDIATE ROCK DIMENSION (INCHES)
D_{15}	11
D_{50}	15
D_{85}	20
D_{100}	30

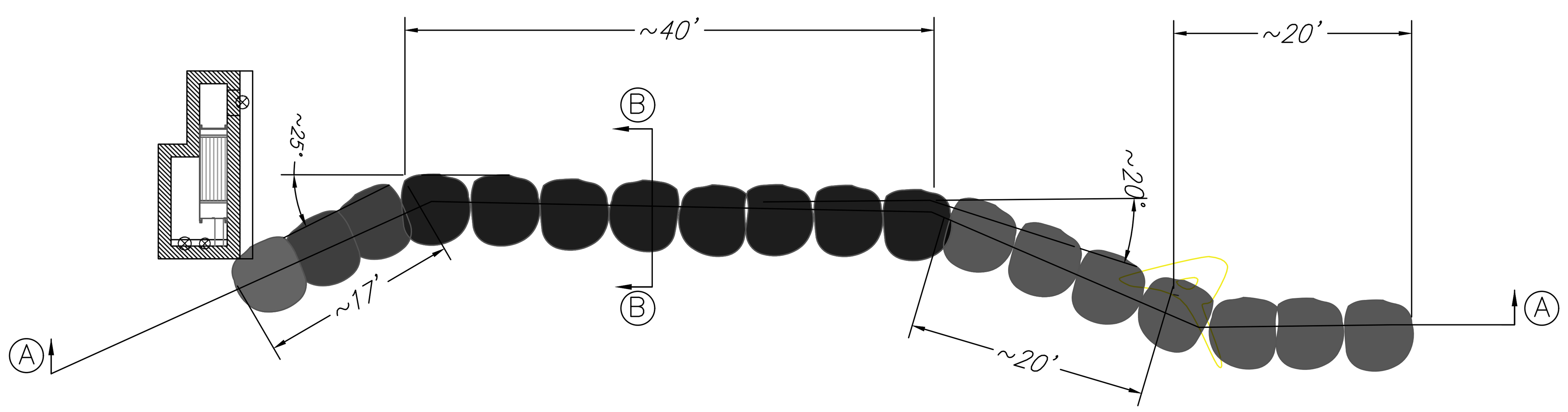
DESIGNED: SBH	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
CADD SBH	C11	RIVER AND WASH PROFILE	PMIS/PKG NO.
TECH. REVIEW: CAH		CRAWFORD DIVERSION FISH PASSAGE ZION NATIONAL PARK	SHEET
DATE: 02/01/26			13 OF 17



ROCK RAMP CREST PLAN VIEW

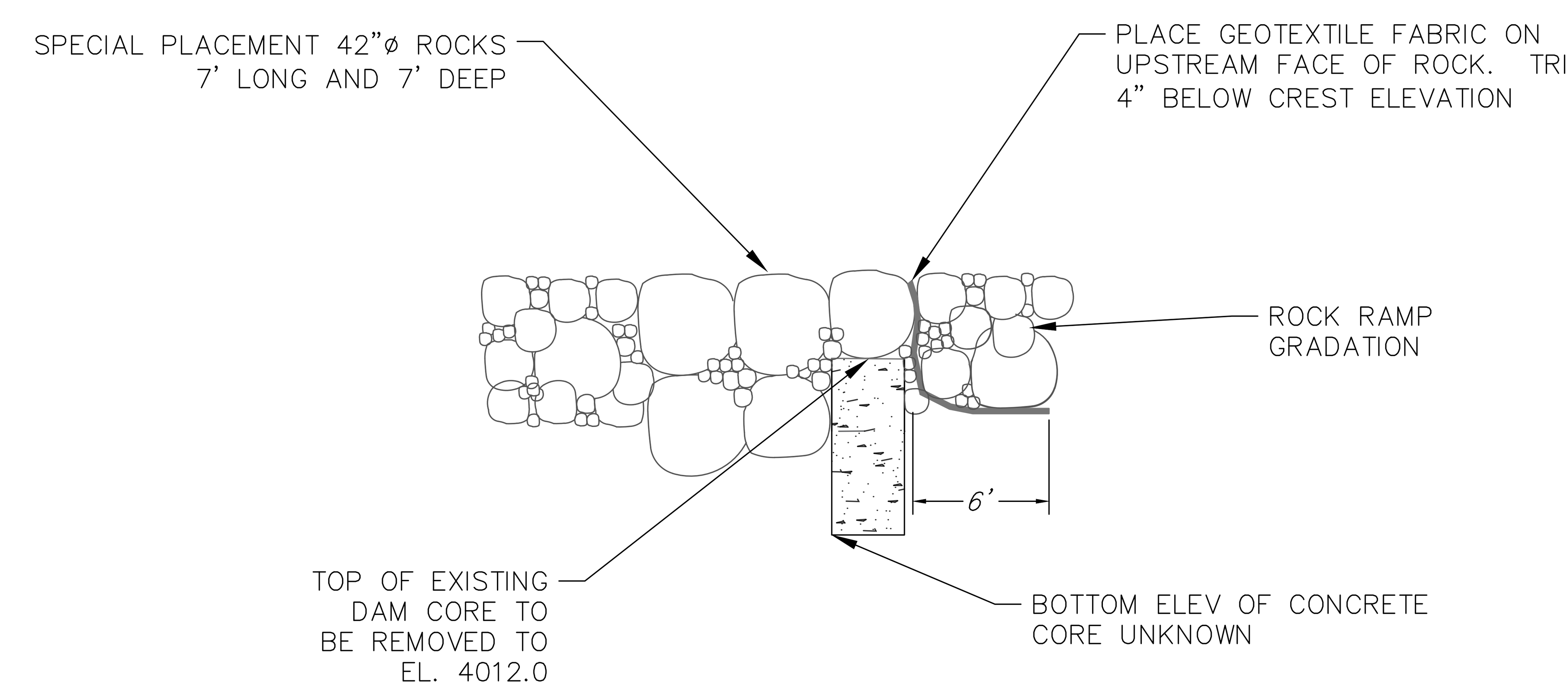
SCALE: 1"=10'
(IF PRINTED 34"X22")
(ROCKS NOT TO SCALE)

LARGE ROCK TO REMAIN



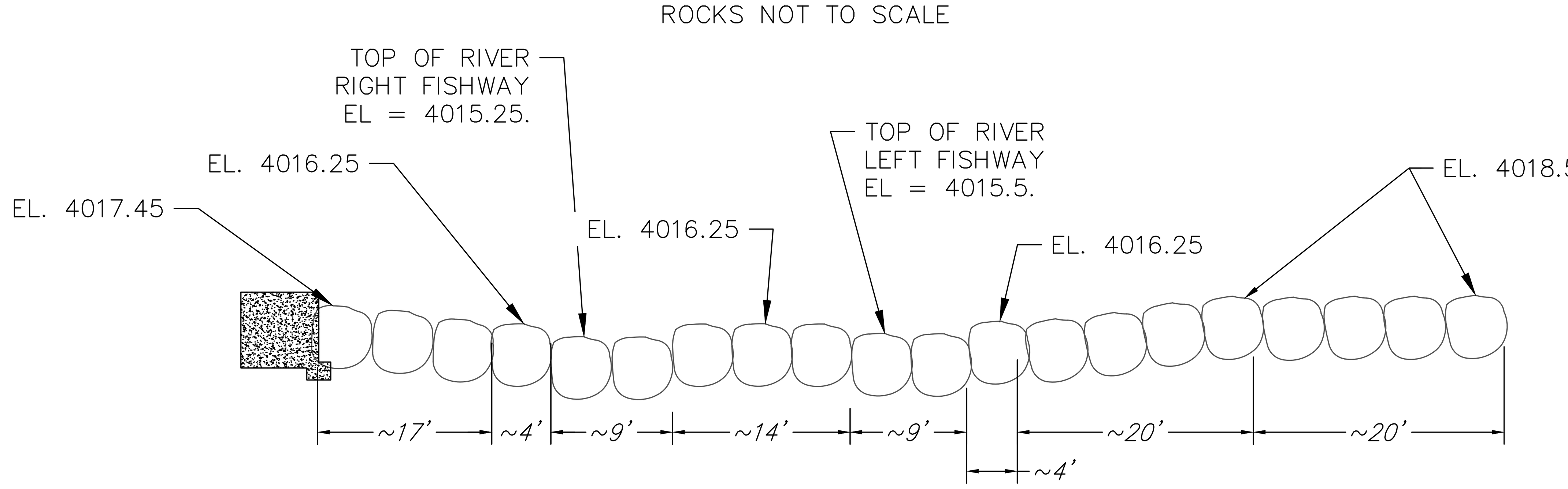
ROCK RAMP CREST PLAN VIEW

SCALE: 1"=10'
(IF PRINTED 34"X22")
ROCKS NOT TO SCALE



SECTION B

SCALE: 1"=5'
(IF PRINTED 34"X22")
ROCKS NOT TO SCALE



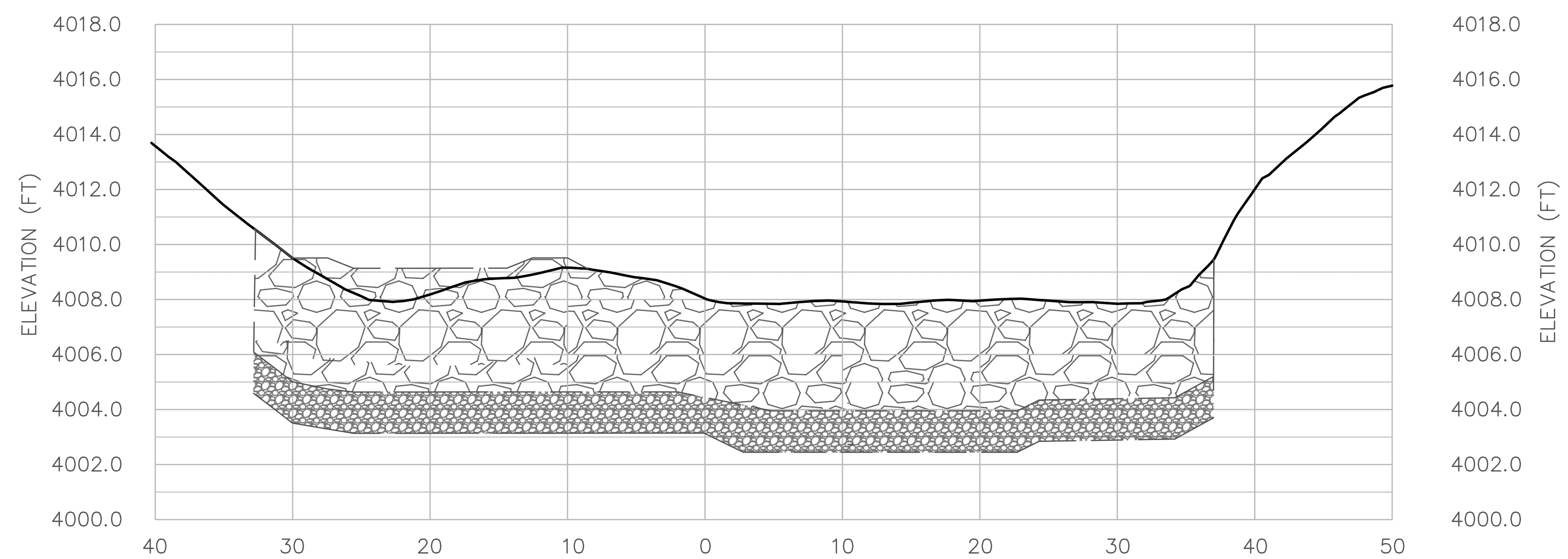
SECTION A

SCALE: 1"=10'
(IF PRINTED 34"X22")
ROCKS NOT TO SCALE

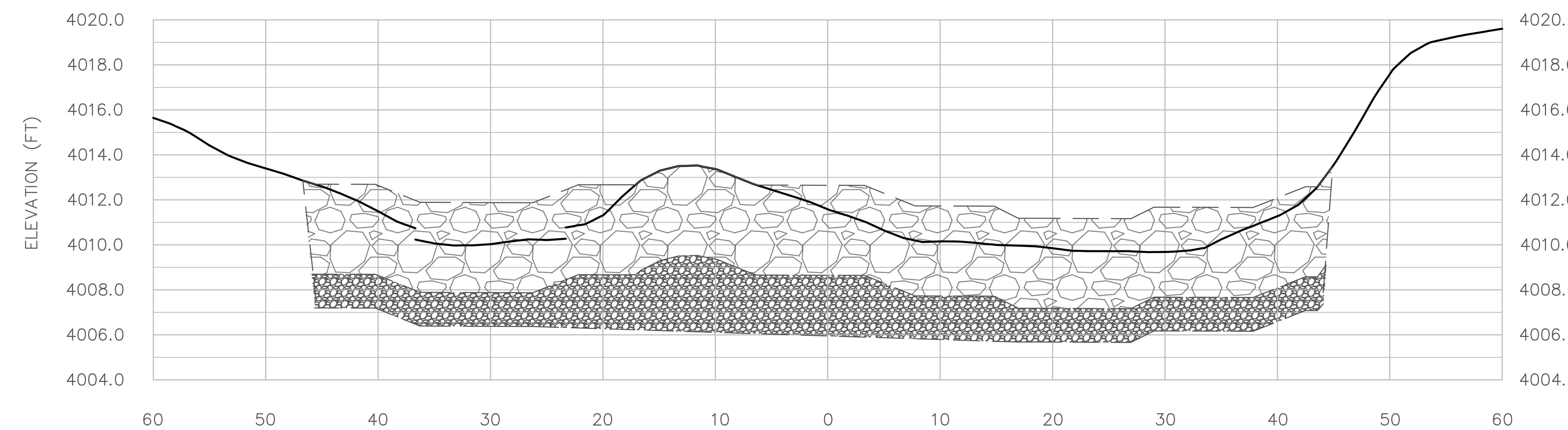
PROFESSIONAL ENGINEER # 11865813-2202 SUZANNE HUHTA STATE OF UTAH 2/1/26

DESIGNED: SBH	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
CADD SBH	C12	ROCK RAMP CREST DETAILS	———
TECH. REVIEW: CAH		CRAWFORD DIVERSION FISH PASSAGE ZION NATIONAL PARK	PMIS/PKG NO.
DATE: 02/01/26			SHEET 14 OF 17

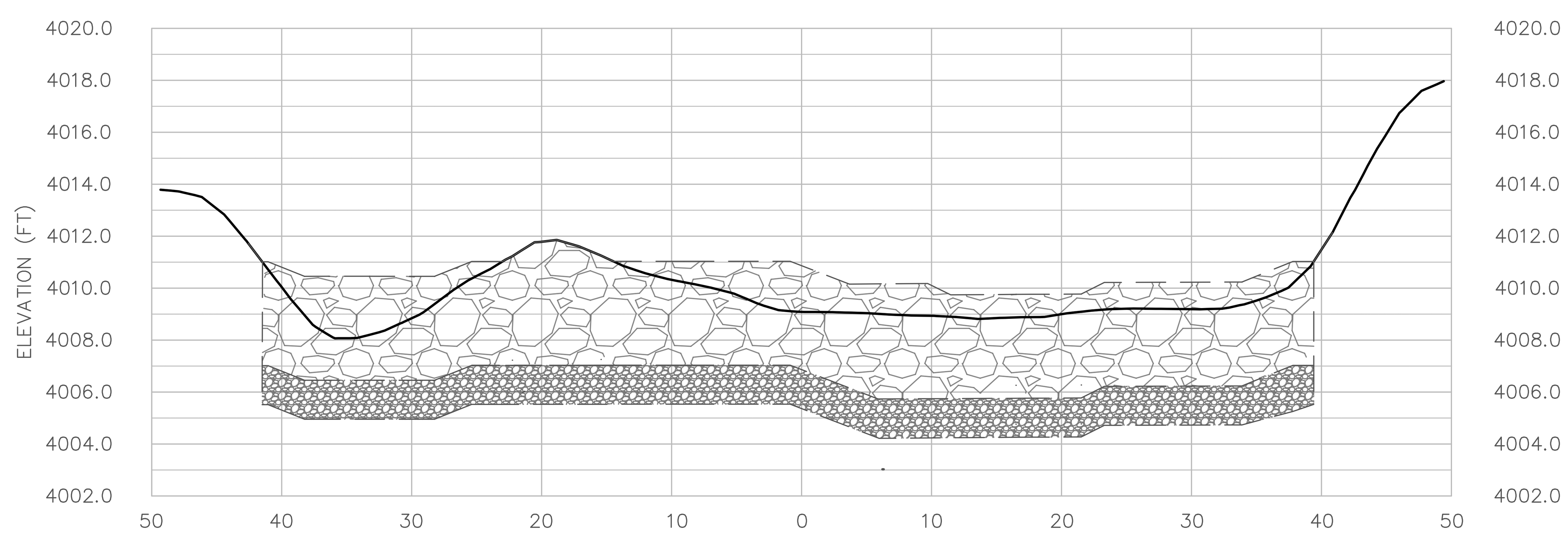
FILE NAME: 2026-02 RR CREST DETAILS C12.DWG
PLOT TIME/DATE: 1:29 PM, 2/1/2026
PLOTTED BY: Suzanne Huhta
PLOT STYLE: Acad.ctb



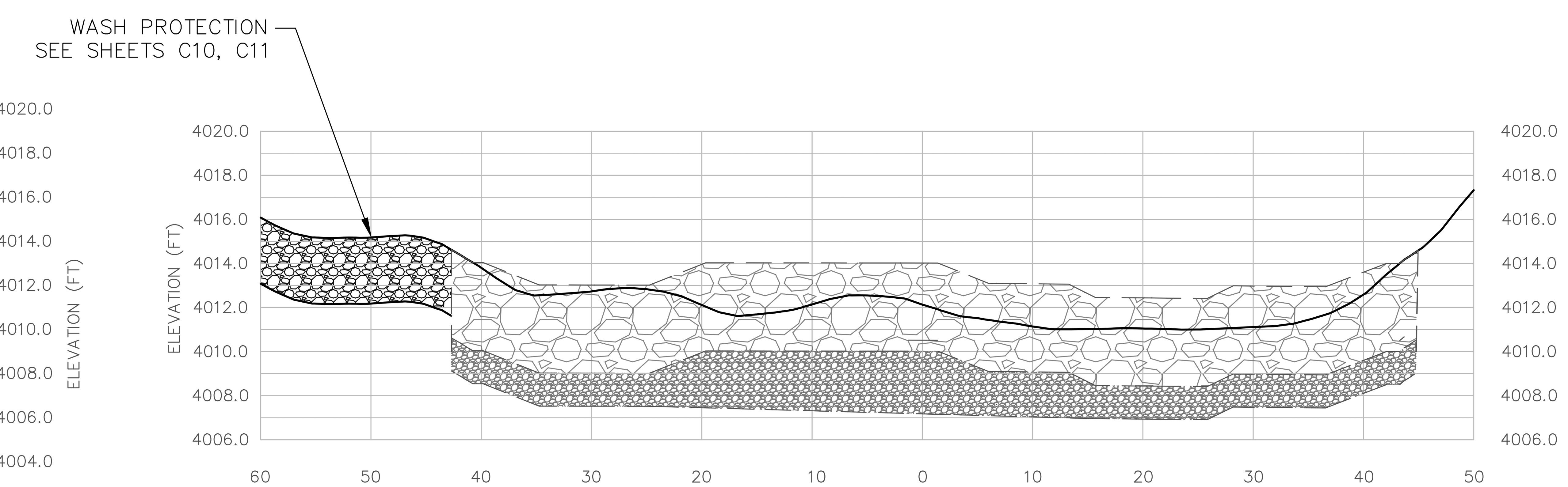
STATION 97+50
CENTERLINE DISTANCE (FT)



STATION 98+50
CENTERLINE DISTANCE (FT)



STATION 98+00
CENTERLINE DISTANCE (FT)



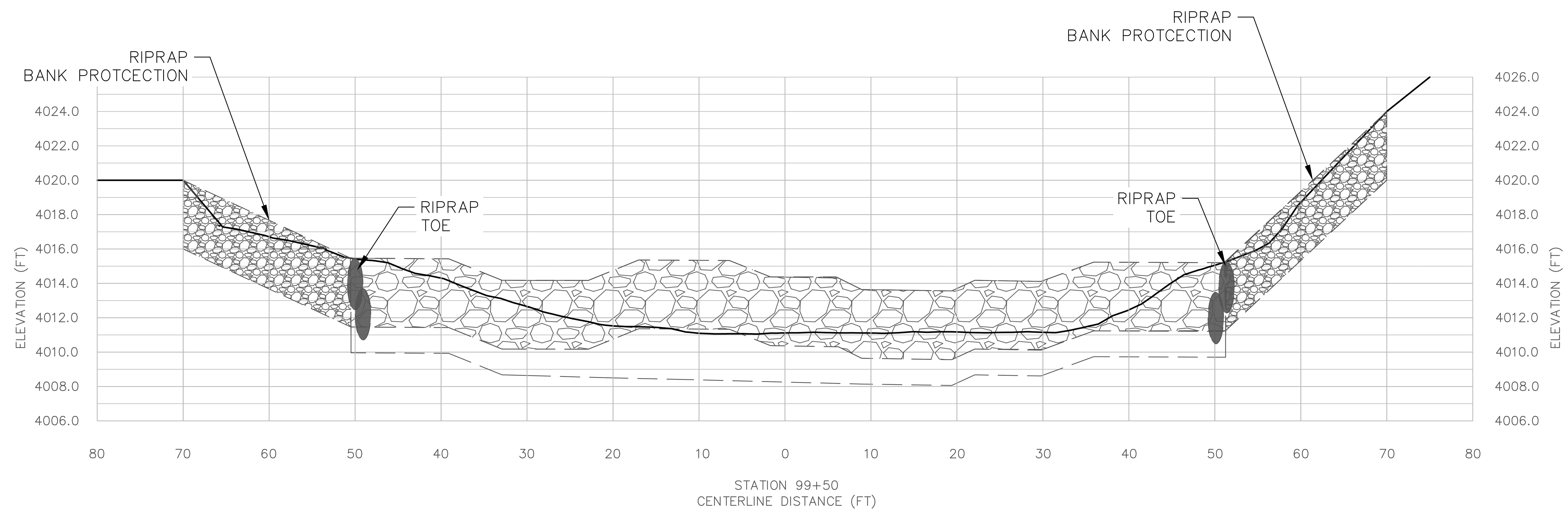
STATION 99+00
CENTERLINE DISTANCE (FT)

NOTES:
1. CROSS SECTIONS ARE SHOWN LOOKING DOWNSTREAM.
2. CROSS SECTIONS ARE ALONG NEW CL.

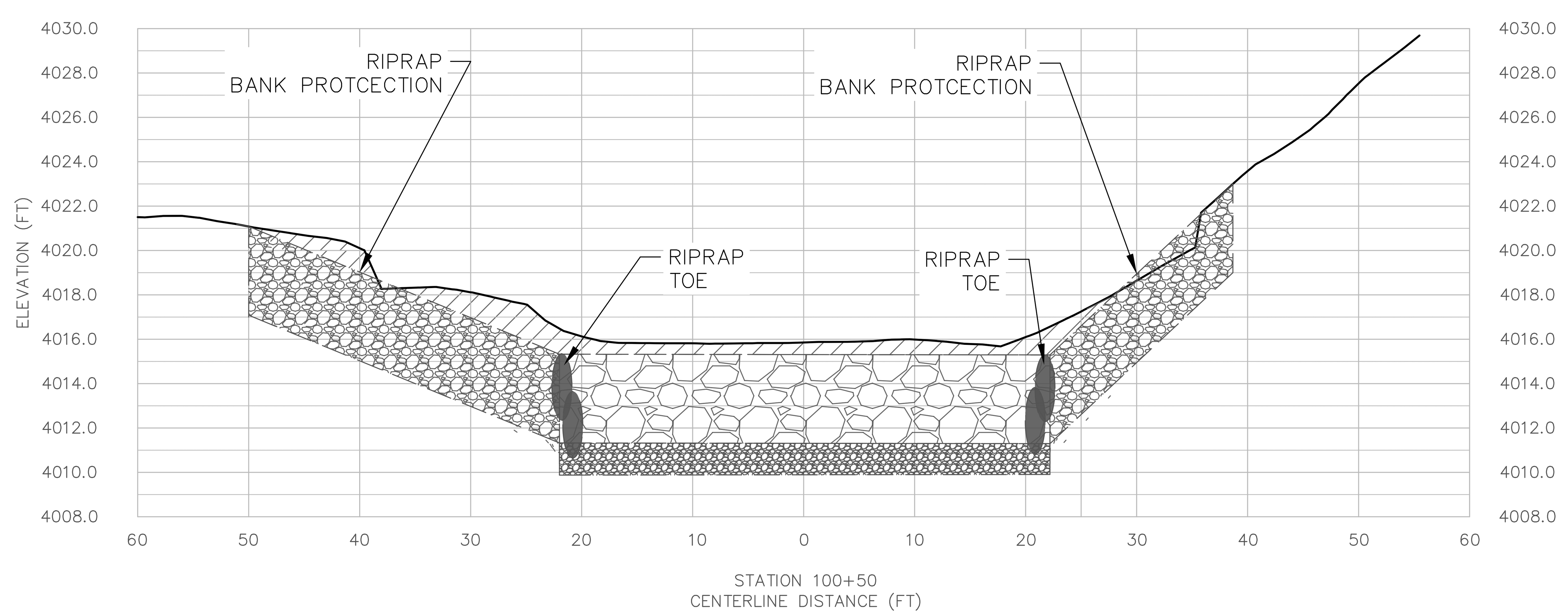
EXISTING GROUND
PROPOSED GROUND
MATERIAL TO BE REMOVED
ROCK RIFFLE
FILTER LAYER

PROFESSIONAL ENGINEER
11865813-2202
SUZANNE HUHTA
STATE OF UTAH

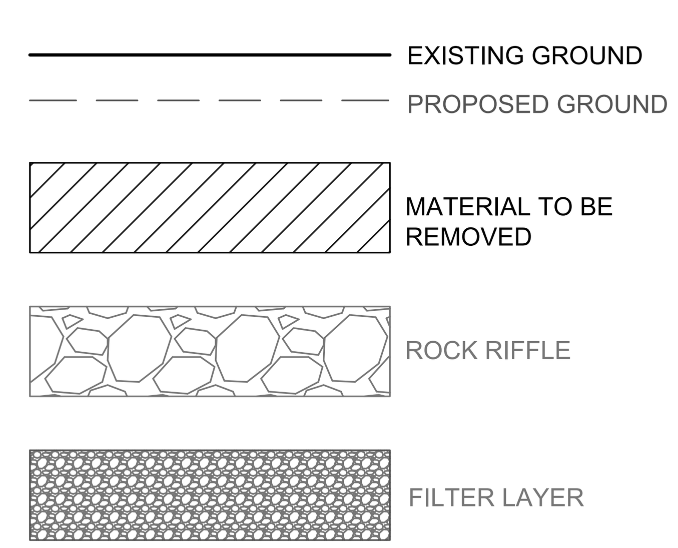
DESIGNED: SBH	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
CADD SBH	C13	RIVER CROSS SECTIONS (1 OF 2)	PMIS/PKG NO.
TECH. REVIEW: CAH		CRAWFORD DIVERSION FISH PASSAGE ZION NATIONAL PARK	SHEET
DATE: 02/01/26			15 OF 17



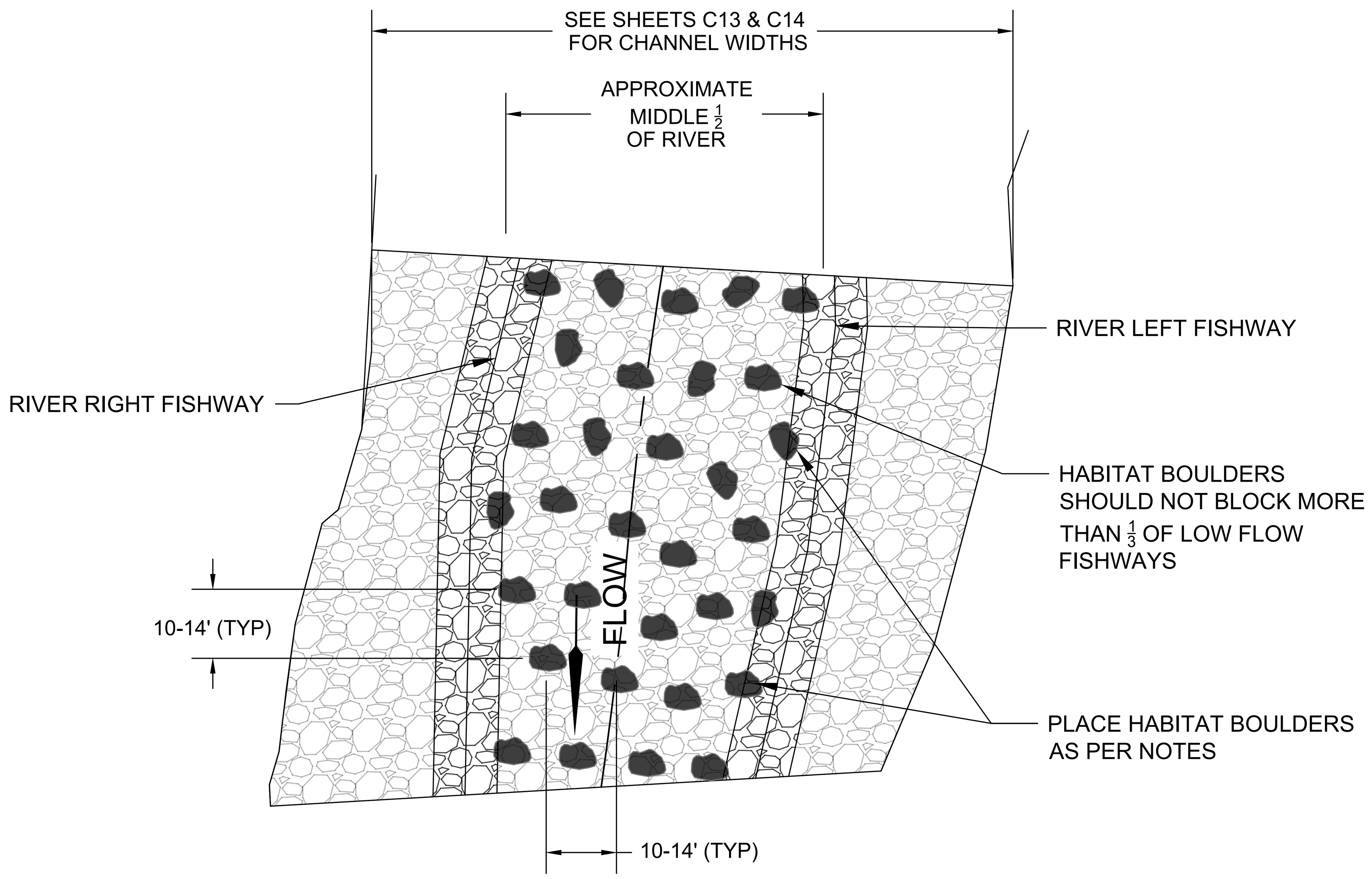
STATION 100+00
SEE SHEET C12



NOTES:
1. CROSS SECTIONS ARE SHOWN LOOKING DOWNSTREAM.
2. CROSS SECTIONS ARE ALONG NEW CL.



DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
SBH	C14	RIVER CROSS SECTIONS (2 OF 2)	———
CADD			PMIS/PKG NO.
SBH			
TECH. REVIEW:			SHEET
CAH		CRAWFORD DIVERSION FISH PASSAGE	16 OF 17
DATE: 02/01/26		ZION NATIONAL PARK	



LARGE ROCK "HABITAT BOULDER" NOTES:

1. A TOTAL OF 110 HABITAT BOULDERS WILL BE INSTALLED.
2. EACH HABITAT BOULDER WILL BE CONSIDERED THE EQUIVALENT OF ONE D_{100} BOULDER.
3. HABITAT BOULDERS 4.25' TO 5.0'ø
4. HABITAT BOULDERS SHOULD PROJECT ABOVE THE GRADE BY 12" TO 18".
5. PLACE BOULDERS IN THE MIDDLE HALF OF THE RIVER.
6. BOULDERS SHALL BE SPACED 10' TO 14' APART IN ALL DIRECTIONS.
7. EXISTING ROCK WITHIN THE LOCATION MAY BE USED WITHOUT RELOCATION WITH C.O. APPROVAL.

GENERAL NOTES:

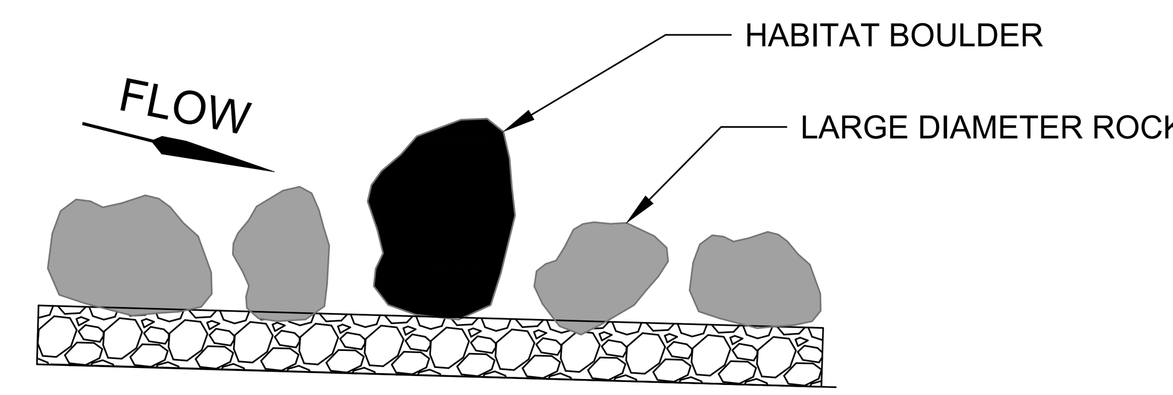
1. SEE SHEET C11 FOR ROCK GRADATIONS.
2. ALL BOULDERS USED IN THIS PROJECT SHALL BE LOCALLY SOURCED BASALT OR SANDSTONE WITH A MOHS HARDNESS > THAN 5.
3. CONTRACTOR IS ENCOURAGED TO RECYCLE ON SITE MATERIALS.

ROCK RAMP PARTIAL PLAN VIEW

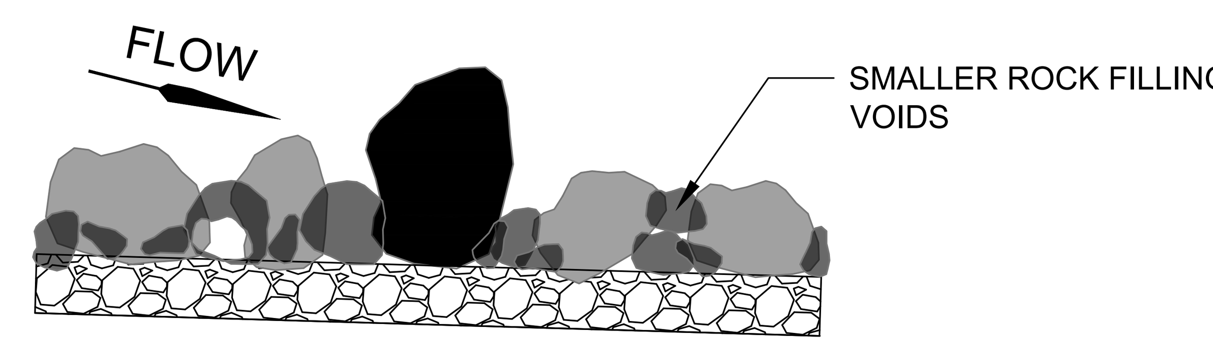
SCALE: N.T.S.

ROCK RAMP NOTES:

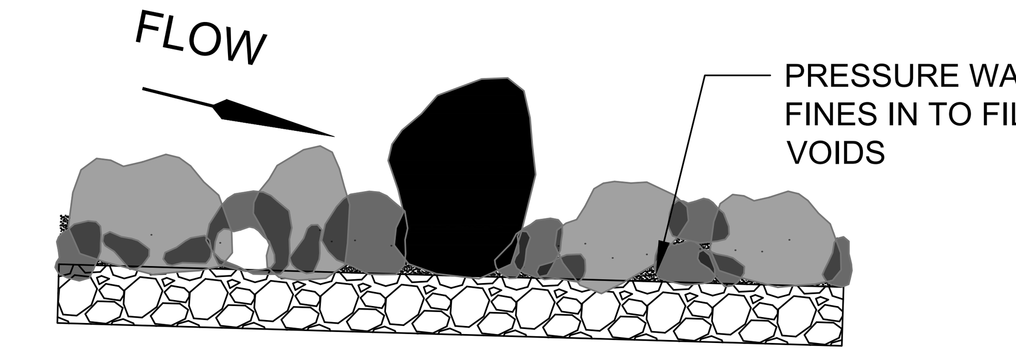
1. ROCK PLACEMENT FOR THE RAMP SHALL GENERALLY OCCUR IN 2 OR MORE LIFTS AS FOLLOWS:



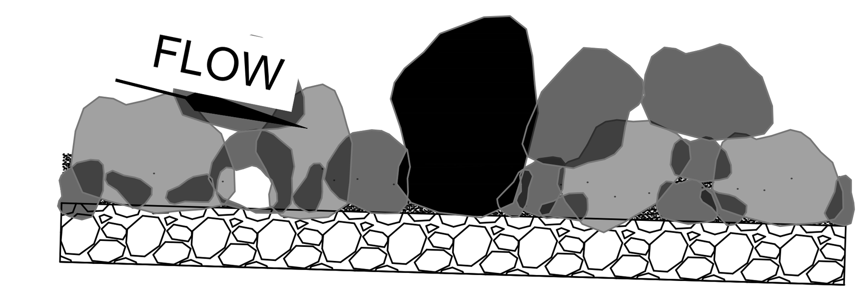
STEPS
- 1.1. PLACE ROCK FILTER LAYER.
- 1.2. PLACE ~½ OF LARGE DIAMETER ROCK (D_{85} AND D_{100}) AND ALL HABITAT BOULDERS.



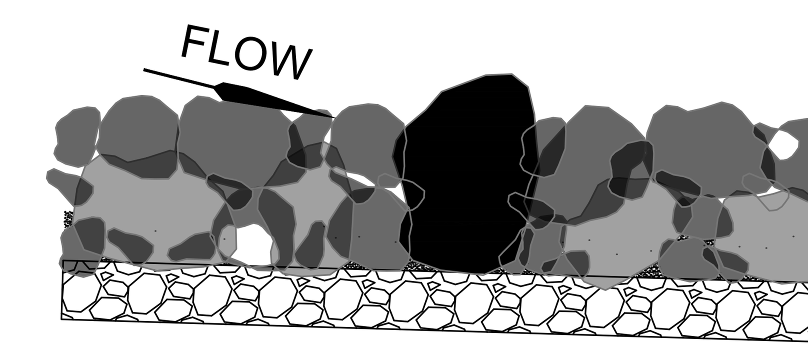
- 1.3 PLACE 18" TO 24" LIFT OF D_{15} THROUGH D_{50} ROCK AROUND LARGE ROCK.
- 1.4 COMPACT TO CREATE INTERLOCKING GRID.



- 1.5 PRESSURE WASH NATIVE MATERIAL <0.5"ø INTO VOID AREAS TO CONSOLIDATE.
- 1.6 FINISHED SURFACE SHALL TEMPORARILY POND WATER OR AS APPROVED BY C.O.



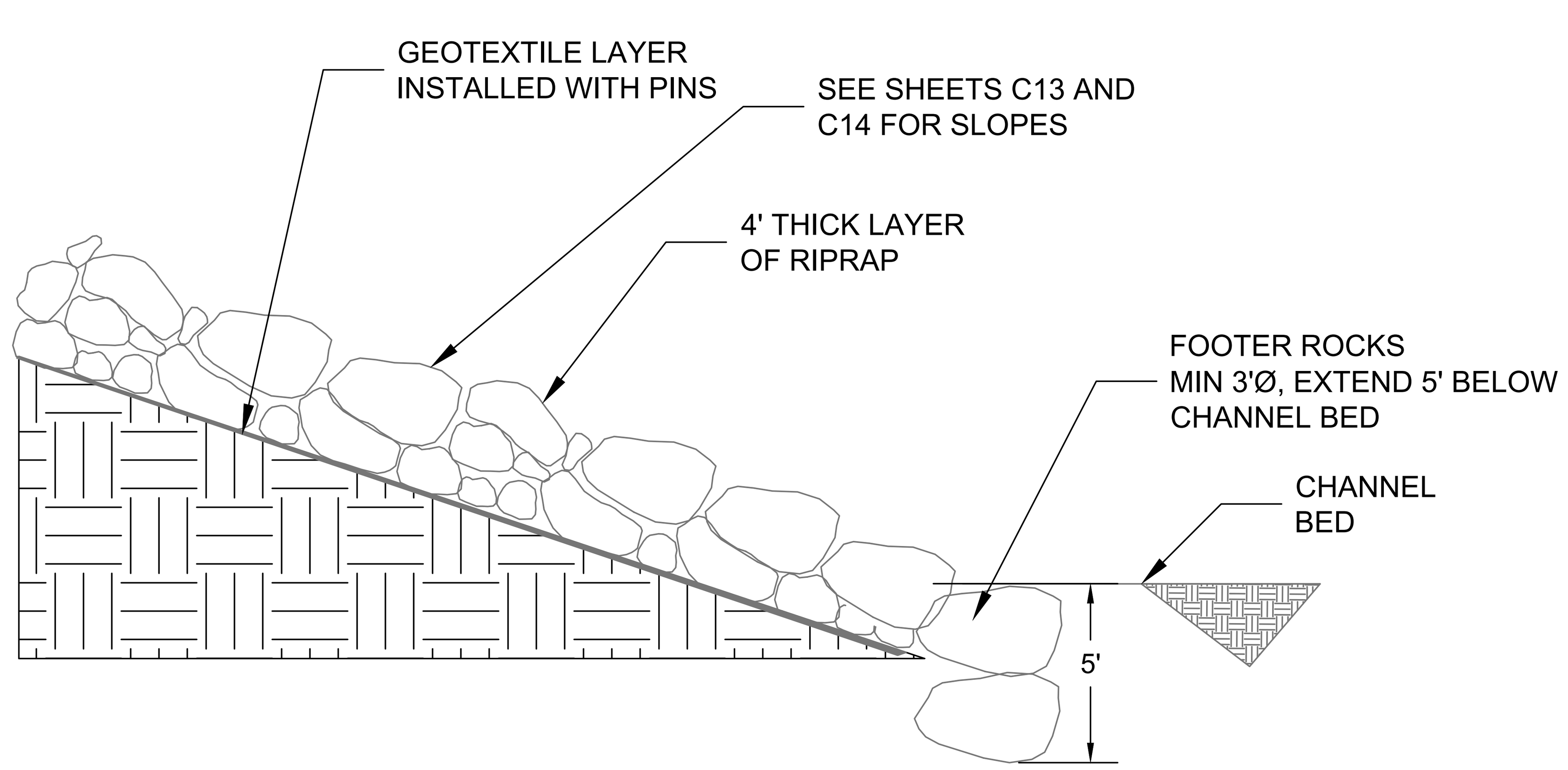
- 1.7 PLACE REMAINING LARGE ROCK.
- 1.8 ~ ½ OF THE TOTAL D_{85} AND D_{100} ROCK SHOULD PROTRUDE BETWEEN 3" AND 6" ABOVE FINISHED GRADE.
- 1.9 HABITAT BOULDERS SHOULD PROTRUDE ABOVE THE GRADE BY 12" TO 18".



- 1.10 PLACE REMAINING ROCK.
- 1.11 PRESSURE WASH SUBSTRATE IN VOID AREAS TO CONSOLIDATE. ADD MATERIAL TO MAKE UP FOR SETTLEMENT. FINISHED SURFACE SHALL TEMPORARILY POND WATER OR AS APPROVED BY C.O.

PARTIAL PROFILE VIEWS AT DIFFERENT CONSTRUCTION STEPS

SCALE: N.T.S.



RIPRAP GRADATION	
% SMALLER THAN GIVEN SIZE BY WEIGHT	INTERMEDIATE ROCK DIMENSION (INCHES)
D_{15}	14.5
D_{50}	24
D_{85}	38
D_{100}	48

RIPRAP DETAIL

SCALE: N.T.S.

RIPRAP NOTES:

1. REFER TO SHEETS C5 AND C6 FOR ACTUAL LOCATION AND LIMITS.
2. MAXIMUM SLOPE OF 2.5H:1V, SEE SHEETS C13 AND C14 FOR SLOPES.
3. 48" RIPRAP THICKNESS MIN.
4. GENERAL PLACEMENT TECHNIQUES SHOULD RESULT IN LARGER ROCK AT THE SURFACE AND ROCK SECURELY INTERLOCKED AT THE DESIGN THICKNESS AND GRADE. COMPACTION AND LEVELING SHOULD RESULT IN MINIMAL VOIDS AND PROJECTIONS ABOVE GRADE.
5. GEOTEXTILE WITH PINS AS PER MANUFACTURER SPECIFICATIONS.
6. SEE SPECIFICATIONS FOR GEOTEXTILE INFORMATION

PROFESSIONAL ENGINEER
11865813-2202
SUZANNE HUHTA
STATE OF UTAH

DESIGNED: SBH	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
CADD SBH	C15	RIVER GRADING DETAILS	———
TECH. REVIEW: CAH		CRAWFORD DIVERSION FISH PASSAGE ZION NATIONAL PARK	PMIS/PKG NO.
DATE: 02/01/26			SHEET 17 OF 17